

Catalogue of Catalogues: A Web Platform for Earthquake Catalogue Management, Merging, and Statistical Analysis

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Abstract

We present the *Catalogue of Catalogues* (CofC), an open-source web platform for ingesting, quality-scoring, merging, and statistically analysing earthquake catalogues from multiple agencies and formats, following the FAIR data principles (Findable, Accessible, Interoperable, Reusable). CofC accepts data as CSV/TXT, GeoJSON, QuakeML 1.2 Basic Event Description (BED), and the FDSN Event Web Service. A six-strategy merging engine associates duplicate entries one to one (each entry with its closest counterpart in every other catalogue) within configurable, adaptive time and distance windows (widened for larger and deeper events), with magnitude and depth consistency enforced as separate per-group gates, and resolves conflicts by solution quality, source priority, most recent solution, record completeness, or an averaged or median epicentre, with optional per-field rules for depth, magnitude and focal mechanism and the option to hold flagged groups for review, keeping every contributing entry as per-event lineage. A transparent quality-scoring system (a 0–100 index with letter grades) evaluates each event across five dimensions: location precision, network geometry, solution fit, magnitude quality, and analyst-evaluation status. Standard seismological analyses run client-side in the browser: Gutenberg-Richter b -value estimation by maximum likelihood, magnitude of completeness (M_c) by b -value stability (the default), a goodness-of-fit test or Maximum Curvature (MAXC), Gardner-Knopoff declustering, and cumulative seismic moment and energy. A fully synthetic two-catalogue New Zealand example (no real event data), processed end to end by the platform’s own engine, illustrates the workflow and quantifies the biases an analyst must control: the matcher pairs 99% of the earthquakes both agencies report, with 98% of its pairs correct; a quality threshold removes 94% of the events with azimuthal gap above 180° but also favours

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larger events, lowering b by 0.06; the MAXC and goodness-of-fit estimates of M_c fall well below the completeness of the merged catalogue, which the more conservative b -value-stability estimate reaches; and window declustering lowers b by a further 0.11. Above a stable cut-off, the estimator recovers the planted $b = 1.00$ of the independent events (1.01 ± 0.01). These values characterise the controlled synthetic inputs, not real New Zealand seismicity. We identify six categories of statistical bias that affect merged-catalogue analysis and give practical mitigation guidance. Source code and documentation are freely available under the MIT licence.

1 Introduction

1.1 The role of earthquake catalogues in seismology

Earthquake catalogues are the observational basis of quantitative seismology. Methods from probabilistic seismic hazard analysis (PSHA; [Cornell 1968](#); e.g. the 2022 Aotearoa New Zealand NSHM, [Gerstenberger et al. 2024](#)) through operational earthquake forecasting ([Schorlemmer et al., 2007](#)) to tectonic geodesy all rest on the assumption that the catalogue faithfully represents the space-time-magnitude distribution of seismicity. The Gutenberg-Richter (GR) frequency-magnitude relation ([Gutenberg and Richter, 1944](#)) and its b -value, which encodes the relative proportion of small to large earthquakes ([Kagan, 1999](#); [Schorlemmer et al., 2005](#)), are estimated directly from catalogue data; any systematic error in the catalogue propagates directly into hazard and risk assessments. Similarly, the Epidemic-Type Aftershock Sequence (ETAS) model ([Ogata, 1988](#)), the most widely used stochastic framework for short-term earthquake forecasting, requires a complete, Poissonian background catalogue that is free from both incompleteness and aftershock contamination.

The same region is often covered by several catalogues from different providers, each built with different station networks, velocity models, magnitude scales, analysis procedures, and update frequencies. Examples include GeoNet (New Zealand; [Petersen et al. 2011](#)), the Northern California Earthquake Data Center ([Northern California Earthquake Data Center, 2014](#)), the International Seismological Centre (ISC; [Storchak et al. 2013](#)), and the Engdahl-van der Hilst-Buland relocated global catalogue ([Engdahl et al., 1998](#)). Researchers routinely need to *merge* or *compare* these sources: to extend a short modern catalogue with historical data, to supplement a global catalogue with a higher-resolution regional one, or to construct a unified input for PSHA over a region monitored by multiple agencies. This multi-source workflow introduces three classes of problem that, if ignored, produce statistically invalid results.

1.2 Problem 1: Format heterogeneity

No universally adopted exchange format exists for seismological data. The QuakeML 1.2 Basic Event Description (BED) schema ([Schorlemmer et al., 2011](#)) provides an XML vocabulary covering origins, magnitudes, focal mechanisms, picks, arrivals, and amplitude measurements, and is the standard output of the FDSN Event Web Service ([International Federation of Digital Seismograph Networks, 2013](#)). However, many agency archives remain in agency-specific CSV layouts, plain text bulletins, or legacy binary formats. The GeoJSON format ([Butler](#)

et al., 2016) has gained traction for web-facing services but provides no controlled vocabulary for seismological fields; column and property names vary freely between providers. When a researcher assembles a multi-source dataset, the practical first step is writing bespoke format-conversion code, a task that is repetitive, error-prone, and rarely shared or version-controlled.

1.3 Problem 2: Duplicate events and location scatter

When two catalogues cover the same region, the same earthquake typically appears in both, reported with slightly different origin times, hypocentres, and magnitudes. These differences arise from physical causes: different station sets give different travel-time residuals; regional versus global velocity models bias depth systematically; and automatic picks differ from manually reviewed ones. The global relocation literature documents that, even for well-recorded teleseismic events, agency-to-agency hypocentre differences commonly reach 10–20 km in epicentre and tens of kilometres in depth, which is far less well constrained (Engdahl et al., 1998). Naïvely concatenating catalogues without duplicate removal counts every earthquake that more than one agency reports once per entry: it inflates $N(M)$ wherever the catalogues overlap, raises the apparent a -value, and distorts rate estimates. Storchak et al. (2013) describe the duplicate-association procedure developed for the ISC-GEM global catalogue, which required multi-year expert review. Automated, configurable duplicate association is available in catalogue-building libraries (GEM Foundation, 2024) and operational monitoring systems (GFZ German Research Centre for Geosciences and gempa GmbH, 2008), but not in a browser-based interface that a regional researcher can apply directly to an arbitrary pair of uploaded catalogues without scripting or an operational deployment.

1.4 Problem 3: Statistical biases in merged catalogues

Even after format conversion and duplicate removal, several statistical biases require active mitigation before standard analyses are valid. We summarise the mechanisms of the four most consequential biases here; the full set of six, their practical mitigation steps, and how CofC supports each are collected in Section 8.

Magnitude of completeness (M_c). A catalogue is complete above M_c , the smallest magnitude at which essentially all events are detected by the network. Below M_c , the GR relation breaks down (Rydelek and Sacks, 1989). Fitting b without first determining and enforcing M_c leads to systematic underestimation (Woessner and Wiemer, 2005). Several methods for estimating M_c exist (Wiemer and Wyss, 2000; Cao and Gao, 2002; Woessner and Wiemer, 2005; Mignan and Woessner, 2012), with different assumptions and biases. When merging two catalogues with different M_c values (e.g., $M_c = 1.5$ for a dense regional network and $M_c = 3.0$ for a global catalogue), the combined FMD will be incomplete between the two thresholds unless the analyst enforces the higher cut-off.

Magnitude scale heterogeneity. Different agencies routinely use different magnitude scales: local magnitude (M_L ; Richter 1935), body-wave magnitude (m_b), surface-wave magnitude (M_s), and moment magnitude (M_w ; Hanks and Kanamori 1979; Kanamori 1977). Each scale saturates at a different value (M_L above ~ 6.5 , m_b above ~ 6.0 ; Bormann and Saul 2009), and inter-scale regressions carry scatter that depends on the regression method (Das et al., 2011; Castellaro et al., 2006). Mixing scales inflates the apparent variance of the FMD,

artificially shifts M_c , and can change the estimated b by up to ± 0.2 (Marzocchi and Sandri, 2003). Tinti and Mulargia (1987) showed that, even with a single scale, binning width and sample size both affect the precision of $\hat{b}$.

Aftershock clustering. Aftershock sequences violate the Poissonian independence assumption underpinning the GR relation and ETAS background estimation. Aftershocks are most numerous at the small magnitudes just above M_c , where they inflate event counts; because clustered aftershock populations are commonly observed to carry a higher b -value than the background, failing to decluster tends to bias the apparent $\hat{b}$ of the raw catalogue *upward* relative to the declustered (Poissonian) catalogue, although the sign and size of the shift depend on the relative size distributions of the clustered and background populations (Marsan and Lengliné, 2008). Declustering itself is not neutral either: window methods remove small independent events more readily than large ones and so bias the declustered $\hat{b}$ low (Mizrahi et al., 2021). We illustrate both shifts directly in Section 7. Several declustering algorithms exist, differing in their sensitivity to swarm-type seismicity and to the aftershocks of aftershocks (van Stiphout et al., 2012): the window method of Gardner and Knopoff (1974), the aftershock-zone scaling of Kagan (2002), the link-based method of Reasenberg (1985), the ETAS-based stochastic method of Zhuang et al. (2002), and the nearest-neighbour method of Zaliapin and Ben-Zion (2013). Helmstetter et al. (2005) demonstrated that ignoring small-magnitude aftershocks, often the majority of any sequence, underestimates cascading stress transfer and therefore the effective rate of background seismicity. The choice of declustering method and parameters is a major source of uncertainty that is rarely propagated through to final hazard estimates.

Location quality filtering. Events recorded by sparse networks have large location uncertainties and poorly constrained depths. If such events are systematically concentrated at one magnitude or depth range, as is typical for offshore events recorded by onshore networks, they introduce spatial bias into b -value and M_c mapping, and can bias an aggregate Gutenberg-Richter fit where the location deficiency correlates with magnitude via detection; a filter on location quality then removes small events preferentially (Section 7). Bondár et al. (2004) showed that network geometry, in particular the *secondary* azimuthal gap, together with the number of stations and the nearest-station distance, controls horizontal location accuracy, with large gaps typically associated with errors of tens of kilometres.

1.5 Existing tools and their limitations

The standard desktop tool for catalogue-based seismicity analysis is ZMAP (Wiemer, 2001), which implements GR analysis, spatial b -value mapping, declustering, and Omori-law fitting in a MATLAB environment. It requires MATLAB licensing and familiarity with its GUI; it can compare two catalogues to flag identical events but does not reconcile them into a merged catalogue, offers no quality-scored event selection, and does not read the QuakeML exchange format. ObsPy (Beyreuther et al., 2010) provides a Python framework for reading and writing QuakeML and performing basic catalogue operations, but all analysis requires custom scripting. EQcorrscan (Chamberlain et al., 2018) enables template-matching and repeating-earthquake detection but produces catalogues that then require separate quality assessment and merging tools. OpenQuake (Pagani et al., 2014) consumes pre-processed catalogues for PSHA but provides no tools for the preprocessing itself. Neither ZMAP,

ObsPy, nor OpenQuake offers a guided workflow for avoiding the biases described above, nor a persistent web-accessible database for storing and sharing processed catalogues. The Community Online Resource for Statistical Seismicity Analysis (CORSSA; [Naylor et al., 2010](#); [Mignan and Woessner, 2012](#); [van Stiphout et al., 2012](#); [Woessner et al., 2010](#)) provides peer-reviewed tutorials on these methods but no software platform.

The need for such integration is intensifying. Template-matching and machine-learning phase-detection workflows now routinely yield catalogues with up to an order of magnitude more events than traditional analyst-reviewed bulletins ([Chamberlain et al., 2018](#); [Mousavi and Beroza, 2022](#); [Illsley-Kemp and Mestel, 2025](#); [Michailos et al., 2025](#); [Chamberlain et al., 2025](#)), while every regional network continues to publish its own authoritative product. A single study area may therefore be covered by several catalogues that differ in completeness, magnitude scale, and location quality and must be reconciled before analysis. This proliferation is spatially concentrated: a companion inventory of New Zealand earthquake catalogues (Figure 1) shows that intensively studied areas, such as the Taupō Volcanic Zone, the northern Hikurangi margin, and the central Alpine Fault, are each covered by many overlapping regional catalogues (panel b), while large offshore tracts, Fiordland, and the far south remain thinly catalogued (panel a). Reconciling these overlapping products into a single, quality-aware view is the task CofC addresses.

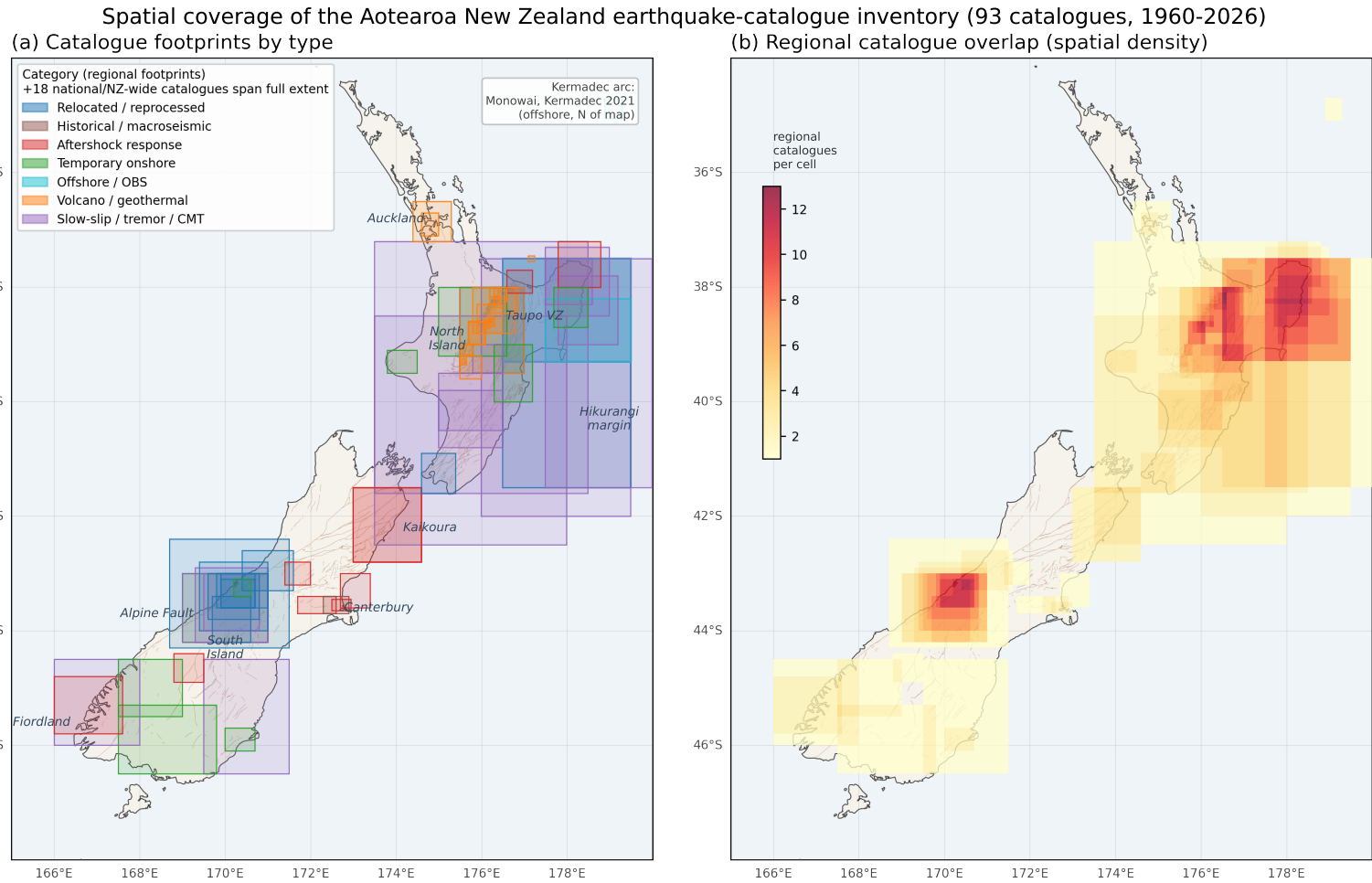


Figure 1: Spatial coverage of a companion inventory of New Zealand earthquake catalogues (93 catalogues, 1960–2026). (a) Footprint of each regional catalogue (a representative bounding box) over the coastline and active faults, coloured by type; 18 national or NZ-wide catalogues span the full extent and are not drawn, and two Kermadec-arc catalogues lie north of the map. (b) Number of regional catalogues overlapping each 0.1° cell, highlighting intensively studied areas (Taupō Volcanic Zone, northern Hikurangi margin, central Alpine Fault) against thinly catalogued offshore tracts, Fiordland, and the far south. This many-catalogues-per-area pattern is the motivation for a merging-and-quality platform.

Format conversion, duplicate association, completeness estimation, and declustering are individually well understood and, in isolation, available in existing tools. The GEM OpenQuake Model Building Toolkit (GEM Foundation, 2024) compiles and homogenises multi-source catalogues with configurable space–time duplicate windows, declustering, and completeness estimation; SeisComP (GFZ German Research Centre for Geosciences and gempa GmbH, 2008) performs automated multi-agency origin association and merging within operational monitoring pipelines. The construction of magnitude-homogeneous catalogues from heterogeneous databases is itself an established methodology (Weatherill et al., 2016), and the PyCSEP toolkit (Savran et al., 2022) provides catalogue processing and statistical forecast evaluation in Python. What is missing is therefore not the algorithms but a single, *browser-native, no-install* platform that chains ingest, configurable merging, transparent per-event quality scoring, and bias-aware analysis behind one interface, preserves provenance at every step, and is usable by a regional researcher on an arbitrary pair of uploaded catalogues without scripting or a configured monitoring deployment. Table 1 contrasts the capabilities of representative tools with those of CofC.

Table 1: Capability comparison of representative catalogue-analysis tools. ✓: native support; (✓): partial support or requires user scripting; –: not supported. CofC’s “Quality scoring” denotes a native but explicit, as-yet-uncalibrated heuristic index (Section 3).

Tool	Multi-format ingest	Duplicate merge	Quality scoring	GR/ M_c / decluster	Web, no install	Provenance/ FAIR export
ZMAP (Wiemer, 2001)	(✓)	–	–	✓	–	–
ObsPy (Beyreuther et al., 2010)	✓	(✓)	(✓)	(✓)	–	(✓)
OpenQuake engine (Pagani et al., 2014)	(✓)	–	–	(✓)	–	–
OQ-MBTK (GEM Foundation, 2024)	✓	✓	(✓)	✓	–	(✓)
SeisComP (GFZ German Research Centre for Geosciences and gempa GmbH, 2008)	(✓)	✓	–	–	–	(✓)
PyCSEP (Savran et al., 2022)	(✓)	–	–	(✓)	–	(✓)
Agency FDSN services	–	–	–	–	✓	(✓)
CofC (this work)	✓	✓	✓	✓	✓	✓

At the institutional level, major providers each publish their own catalogues through FDSN-compliant web services: GeoNet (Petersen et al., 2011), the USGS Comprehensive Catalogue (ComCat; U.S. Geological Survey, Earthquake Hazards Program, 2017), the Swiss Seismological Service, the ISC (Storchak et al., 2013), and the EMSC. The EPOS Seismology Thematic Core Service coordinates access across European networks. ComCat additionally provides a web search interface and the `libcomcat` client for programmatic multi-source retrieval. However, these services expose individual (or centrally pre-merged) agency products rather than providing tools for users to merge, quality-score, and analyse combinations of arbitrary uploaded catalogues from multiple sources. Automated multi-agency event association has been studied for global catalogues (Yeck et al., 2019), as has Bayesian magnitude merging across networks (Si et al., 2024), but these methods are not available as general-purpose tools for regional researchers. In New Zealand specifically, Warren-Smith et al. (2025) conducted the first systematic quantitative assessment of GeoNet location quality using azimuthal gap and uncertainty benchmarks; CofC’s quality index (Section 3) scores the same kinds of quantities, and that assessment is a natural benchmark against which to calibrate it (Section 10). A high-resolution research catalogue for the Taupō Volcanic Zone (Illsley-Kemp and Mestel, 2025) is an example of a specialised regional product that a researcher would want to ingest alongside the national catalogue.

1.6 FAIR principles and reproducibility

A second motivation for CofC is the move towards FAIR scientific data: Findable, Accessible, Interoperable, and Reusable (Wilkinson et al., 2016). Wilkinson et al. (2016) codified these principles to address a crisis of reproducibility in data-intensive science, the same crisis that afflicts earthquake catalogue analysis when preprocessing steps are undocumented, format conversions are ad hoc, and processed datasets are not versioned or persistently identified. The FAIR principles have since been adopted as the basis for data infrastructure by the European Plate Observing System (EPOS), which coordinates seismological data services across 24 European nations. In Aotearoa New Zealand specifically, FAIR is applied alongside the CARE Principles for Indigenous Data Governance (Collective benefit, Authority to control, Responsibility, and Ethics; Carroll et al. 2020), which recognise Māori data-sovereignty interests in nationally significant datasets; the governance and access-control implications are developed in the companion white paper (Graham et al., 2026).

CofC applies the FAIR principles to earthquake catalogue data:

- **Findable.** Every catalogue stored in CofC receives a stable internal identifier and is indexed by searchable catalogue-level metadata (name, institution, temporal and geographic coverage, magnitude range, event count, licence). Globally unique persistent identifiers are not minted by the platform itself; a released catalogue is instead designed to be deposited with an external agency (e.g. DataCite or Zenodo) to obtain a citable DOI (see Versioning, Section 2).
- **Accessible.** A REST API provides machine-readable access to catalogue metadata and event data in multiple formats (CSV, JSON, GeoJSON, KML, QuakeML) under role-based access control: catalogue listings and metadata can be browsed without signing in, while event data and exports require an account with the Viewer, Editor, or Administrator role; the intended national access model is detailed in the companion white paper (Graham et al., 2026).
- **Interoperable.** Data are normalised to a canonical internal schema and re-exportable in standard formats. Event types, evaluation modes and evaluation statuses are stored in the QuakeML 1.2 BED enumerations (Schorlemmer et al., 2011); magnitude types, for which QuakeML defines no controlled vocabulary, are kept as reported, since their case is meaningful (m_B is not m_b).
- **Reusable.** Every merged event keeps its lineage (the source catalogues and every contributing entry, the merge strategy and parameters, and the quality score of the published solution), and every export carries it together with the catalogue’s version and a SHA-256 checksum of the exported rows, so that a study can state, and a reader can verify, the data state it used.

The design of CofC was shaped by two national community workshops in Aotearoa New Zealand (GSNZ 2023 and GSNZ 2024; Geoscience Society of New Zealand 2023, 2024) that brought together researchers from GNS Science, Victoria University of Wellington, University of Canterbury, and industry. Workshop feedback informed the metadata schema, merge configuration options, quality-grading approach, and access-control model. A companion

white paper (Graham et al., 2026) provides the full FAIR framework specification for the Aotearoa New Zealand national earthquake catalogue infrastructure, including governance, licensing, metadata standards, and role-based access control; the present paper focuses on the software implementation and statistical-analysis workflow.

1.7 The Catalogue of Catalogues

CofC addresses these three catalogue-analysis problems, together with the cross-cutting need for event-level quality assessment, in a single, browser-accessible platform. It ingests heterogeneous formats without custom scripting, detects and removes duplicate events using configurable thresholds grounded in the association standards of Storchak et al. (2013) and Engdahl et al. (1998), scores every event on a transparent quality index, and presents GR analysis and declustering in a guided workflow that surfaces the statistical assumptions at each step. Built on Next.js (Vercel Inc., 2024) and MongoDB (MongoDB Inc., 2024), it requires no local installation and is deployable to any cloud provider or on-premises server. The remainder of this paper describes the platform architecture (Section 2), the merging engine (Section 4), the built-in statistical analyses (Section 5), a worked New Zealand example (Section 7), and detailed guidance on avoiding the most consequential statistical biases (Section 8).

1.8 Contributions

The individual algorithms CofC employs are deliberately standard: maximum-likelihood b -value estimation (Aki, 1965; Utsu, 1965), completeness estimation by b -value stability (Cao and Gao, 2002; Woessner and Wiemer, 2005), goodness of fit or maximum curvature (Wiemer and Wyss, 2000), and Gardner-Knopoff (Gardner and Knopoff, 1974) and Reasenber (Reasenber, 1985) declustering. The contribution of this work is therefore not a new estimator but the *integration* of the multi-catalogue workflow into a single open, web-native, provenance-preserving platform, together with a transparent event-level quality index. Specifically, the contributions are:

1. **An integrated, browser-based workflow** spanning heterogeneous-format ingest, configurable duplicate detection, quality scoring, conflict-resolution merging, and standard seismicity analysis, requiring no local installation or licensed software (Table 1). Catalogue-building libraries such as the GEM toolkit (GEM Foundation, 2024) and operational systems such as SeisComP (GFZ German Research Centre for Geosciences and gempa GmbH, 2008) cover much of the analytical chain, but to our knowledge none delivers it through a browser-native, no-install interface with an integrated per-event quality index.
2. **A transparent, configurable event-level quality index** (equation (1)): a five-dimension 0–100 score with letter grades. Its components (location precision, network geometry, solution fit, magnitude quality, and evaluation status) are the standard determinants of solution quality used in agency review and ground-truth location-quality classification (Bondár et al., 2004; Woessner and Wiemer, 2005; Warren-Smith et al., 2025); the contribution is their combination into a single reproducible, user-configurable index that is computed on import and exposed throughout a public web platform, rather than a new quality criterion.

3. **Configurable, automated duplicate detection and conflict resolution** for arbitrary catalogue pairs, with six interchangeable merge strategies and physical-validity gates, generalising the expert-driven association procedures developed for global catalogues (Storchak et al., 2013; Engdahl et al., 1998; Yeck et al., 2019) into a tool regional researchers can apply themselves.
4. **FAIR-by-design provenance**: every merged catalogue records its full per-event lineage (source-catalogue IDs, every contributing entry and the one whose solution was published, merge strategy and thresholds, quality score, review status), re-exported as columns or properties in the CSV, JSON, GeoJSON and KML outputs and as a per-event QuakeML comment (Schorlemmer et al., 2011), with the catalogue version and a checksum of the exported rows recorded in every export, operationalising the FAIR principles (Wilkinson et al., 2016) for routine catalogue work.
5. **A bias-aware, guided analysis path** that surfaces the underlying statistical assumptions (completeness, magnitude-scale homogeneity, declustering, and quality filtering) at the point of use, consolidating guidance previously dispersed across the CORSSA tutorials (Naylor et al., 2010; Mignan and Woessner, 2012; van Stiphout et al., 2012) into an executable workflow (Section 8).

2 Platform Architecture

CofC presents the catalogue workflow as a single guided pipeline (Figure 2): heterogeneous inputs are ingested and normalised, every event is quality-scored on import, duplicate events are detected and resolved during merging, standard seismicity analyses are run in the browser, and processed catalogues are re-exported in standard formats, with all intermediate state and provenance persisted in the database at every step.

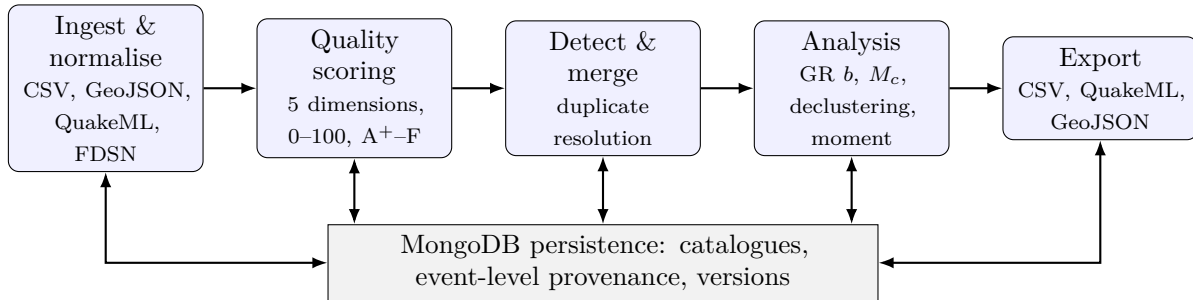


Figure 2: The CofC processing pipeline. Heterogeneous catalogues are ingested and normalised to a canonical schema, quality-scored per event on import (Section 3), merged with configurable duplicate detection (Section 4), analysed in the browser (Section 5), and re-exported in standard formats (Section 6). All stages read and write a MongoDB store that retains catalogues, event-level provenance, and version metadata.

2.1 Technology stack

CofC follows a three-tier architecture: a React 18 / Next.js 15 ([Vercel Inc., 2024](#)) frontend served via the App Router, a serverless backend layer of Next.js API routes, and a MongoDB ([MongoDB Inc., 2024](#)) persistence tier (server ≥ 6 ; accessed through the v7 Node.js driver). All tiers communicate through a versioned REST API; no separate application server is required.

- **Frontend:** React 18, TypeScript, Tailwind CSS, and a component library of 40+ shadcn/ui primitives for forms, tables, dialogs, and data-display cards. Apache ECharts ([The Apache Software Foundation, 2024](#)) provides interactive time-series panels; Leaflet ([Agafonkin, 2024](#)) provides the map layer.
- **Backend:** Next.js API routes deployed as serverless functions, enabling horizontal scaling without managing long-running processes. Events are stored with indexed coordinate fields supporting bounding-box queries, including boxes that cross the antimeridian.
- **Parsers:** Custom TypeScript parsers for CSV/TXT, GeoJSON ([Butler et al., 2016](#)), and QuakeML 1.2 BED ([Schorlemmer et al., 2011](#)). FDSN web-service integration imports live GeoNet data (chunked REST retrieval with automatic time-window subdivision for large queries) into a new catalogue, or into an existing GeoNet-import catalogue, where events are matched by their GeoNet event ID and only those whose fields changed are updated. Records the agency flags as duplicate, not existing, or not locatable are excluded and counted; the agency’s own event type is kept alongside the QuakeML one.

The application is containerised with Docker (Compose recipes for development and production) and is designed to deploy to any cloud provider or on-premises server.

2.2 Data model

Each *catalogue* is a named container with metadata (name, description, source agency, status, and a server-managed version). Each *event* record stores the mandatory seismological fields (origin time, latitude, longitude, depth, magnitude) plus about fifty optional fields drawn from QuakeML: uncertainty estimates (including the error ellipse and its confidence level), azimuthal gap, station and phase counts, magnitude type, evaluation mode and status, agency ID, focal mechanisms, phase picks, arrivals, and amplitude measurements. All QuakeML complex sub-objects are stored as JSON strings so that no information is discarded on import. Every stored event also carries its quality score and grade (Section 3), computed when it is written, and every merged event its lineage (Section 4).

Event types, evaluation modes, and evaluation statuses are stored in the QuakeML 1.2 BED enumerations (a few common non-BED event types, such as volcanic tremor, are kept and mapped to BED types on export), so that these fields are comparable across catalogues. Magnitude types are stored as reported: QuakeML defines no controlled vocabulary for them, and their case carries meaning.

2.3 Versioning and provenance

CofC adopts a three-tier versioning convention (MAJOR.MINOR.PATCH) following semantic-versioning conventions and DataCite data-versioning best practice ([DataCite Metadata Working Group, 2021](#)). The platform manages the version itself: a catalogue starts at 1.0.0 when first loaded, and each operation on it releases one new version, with the time of the change recorded, according to the following policy:

- **Major** increments when event parameters of existing events change (an edited event, or a GeoNet re-import that updates existing events).
- **Minor** increments when events are added to a catalogue that already held events, without changing existing solutions.
- **Patch** increments for catalogue name or metadata corrections that do not affect event parameters.

Recomputing derived fields such as the quality score does not release a version. The version label a depositor gives the source dataset is kept separately. Every export records the catalogue’s version and a checksum of the exported rows (Section 6), so a study can cite the version it used and a reader can verify a copy against it. The platform does not keep the event rows of past versions, so a superseded version cannot be retrieved from it: each major and minor release is intended to be deposited with an external archive (e.g. Zenodo) to obtain a persistent DOI, and DOI minting is not performed by the platform itself. Every merged catalogue carries event-level provenance: the source catalogue IDs, every contributing entry (with the one whose solution was published marked), the merge strategy and threshold parameters applied, the quality score of the published solution and, for a group held back for review, its review status. This per-event provenance is carried in the CSV, JSON, GeoJSON and KML exports and as a per-event QuakeML `comment` ([Schorlemmer et al., 2011](#)), and merge-level metadata, among it the merge’s quality-control summary (Section 4), is recorded at the catalogue level, supporting the reproducibility goals of CSEP-based forecast evaluations ([Schorlemmer et al., 2007](#)).

2.4 Visualisation

CofC provides the following interactive views, on each catalogue’s map page and on the Analysis page, where every view reflects the same event filters.

Interactive map. The map view is built on Leaflet ([Agafonkin, 2024](#)) and renders events as circles whose size steps up with magnitude and whose colour can encode focal depth, quality grade, azimuthal gap, or source catalogue (one colour per contributing catalogue, a merged event taking the colour of the catalogue whose solution it published), and on the Analysis page also magnitude; the legend is generated from the same colouring. Events are drawn on a single canvas layer (Leaflet `preferCanvas`) and, for very large catalogues, sub-sampled to maintain browser responsiveness. The sample is *spatial*: the visible extent is divided into screen cells and the largest event in each occupied cell is retained as its representative, any remaining capacity being filled evenly through the catalogue. The plotted set is therefore deliberately not proportional to the catalogue’s magnitude–frequency distribution, and neither

the symbol density nor the magnitude mix should be read quantitatively. On the Analysis page two overlays, off by default, can be switched on: (i) *uncertainty ellipses*, which draw each plotted event’s horizontal location uncertainty from the agency’s reported error ellipse where there is one, else from its circular horizontal uncertainty, else from the latitude and longitude uncertainties (a circle of the semi-major axis when no orientation is reported), stating the ellipse’s confidence level when the source recorded it; and (ii) *focal-mechanism beach balls*, lower-hemisphere diagrams drawn from each event’s preferred focal mechanism. Both act on the plotted events only and are limited to the 150 largest of them, with a notice when the limit applies.

Station coverage. Station coverage is shown through the per-event azimuthal gap, which is the coverage measure the catalogues report (no station coordinates are stored per event): in the azimuthal-gap colour mode the map colours events on a continuous ramp from green (small gap) through orange to red at 180°, continuing into magenta and violet beyond 180° so that poorly constrained events stand out, with events that report no gap in grey. It shows in map view what a gap distribution such as Figure 5a shows in aggregate, so that regions where network geometry is insufficient for reliable locations stand out spatially.

Frequency-magnitude distribution. The Gutenberg-Richter and M_c tabs plot the cumulative and non-cumulative event counts by magnitude bin and overlay the fitted GR line and the M_c estimate; under the frequency-magnitude distribution the M_c tab plots the b -value stability test at every cut-off, whichever method set M_c (Section 5.2). When the magnitude filter has a lower bound, the fit uses it as an explicit cut-off (and says so); the filter’s upper bound is never applied to the fit, which needs the distribution’s upper tail, and M_c itself is always estimated from the untruncated distribution, with the method and correction chosen in the M_c settings (Section 5.2).

Time-series panels. The Temporal tab shows a seismicity-rate series, the cumulative event count, magnitude against origin time (with the M_c or cut-off line; above 3 000 events a sample made of the largest events plus an even stride in time), and the cumulative release of seismic moment or radiated energy. The rate series counts the events at or above the magnitude cut-off, or above the estimated M_c when no cut-off is set, per bin of one UTC day, ISO week or calendar month (by default daily for spans up to a year and weekly beyond), so that step changes in network completeness show up as rate anomalies. A first or last bin that the analysed period covers only in part is drawn dashed. It is scaled to a full bin when that period is known: the time filter’s window and, when a single catalogue is analysed, the catalogue’s declared time period, intersected when both are set (the end is exclusive, and a date-only end covers that day). Otherwise its coverage can only be read off the first or last event, so it is drawn at its raw count, with a note that it may understate the rate. Cumulative curves are plotted at the end of each bin, and events whose origin time cannot be parsed are left out of the series and counted. The Analysis page withholds the moment and energy totals, like the b -value and M_c , for a set pooled from several catalogues, which would count shared earthquakes more than once.

2.5 Supported input formats

Table 2 summarises the four supported input paths.

Table 2: Input formats supported by CofC.

Format	Extension(s)	Notes
CSV / TXT	.csv, .txt, .dat	Columns resolved by an alias table, with date order, depth unit and longitude convention decided per file; the mapping step shows the resolution and applies only the user’s explicit changes.
GeoJSON	.geojson, .json	RFC 7946 (Butler et al., 2016) Feature or FeatureCollection; properties auto-mapped.
QuakeML 1.2 BED	.xml, .qml, .quakeml	Full preferred-origin / preferred-magnitude extraction; all nested arrays preserved.
GeoNet live import	(FDSN WS)	Configurable date range and region; into a new catalogue or an update of an existing GeoNet import.

3 Quality Scoring and Validation

CofC computes an event-level quality index on import and makes it available to every downstream step: merging, filtering, and analysis. This section defines that index and the companion cross-field consistency checks.

3.1 Event-level quality index

Upon import every event receives a quality score $Q \in [0, 100]$ computed as a weighted sum over five dimensions of solution quality:

$$Q = w_{\text{loc}}Q_{\text{loc}} + w_{\text{net}}Q_{\text{net}} + w_{\text{sol}}Q_{\text{sol}} + w_{\text{mag}}Q_{\text{mag}} + w_{\text{eval}}Q_{\text{eval}}, \quad (1)$$

with default weights $w_{\text{loc}} = 0.35$, $w_{\text{net}} = 0.25$, $w_{\text{sol}} = 0.15$, $w_{\text{mag}} = 0.15$, $w_{\text{eval}} = 0.10$. The weights are largest for the dimensions that most directly govern whether a solution is usable in quantitative analysis: location precision and network geometry are the primary determinants of hypocentral accuracy (Bondár et al., 2004; Warren-Smith et al., 2025); the arrival-time fit and magnitude quality are secondary; and the analyst-evaluation status, informative but partly redundant with the instrumental metrics, carries the least weight. Each dimension applies an explicit penalty when its underlying fields are absent, so events with sparse metadata are scored conservatively. The weights are intentionally round-valued and are exposed as configurable parameters of the scoring module (defaults above), so that communities with different priorities can adjust them without modifying the analysis code. Because Q_{loc} and Q_{net} are functions of network density, Q is positively correlated with network coverage by construction: it is an explicit, reproducible ranking rather than a network-neutral measure, and the default weights are a design choice not yet calibrated against an independent ground-truth benchmark (Section 10). Field completeness, physical-consistency checks, and magnitude-scale

homogeneity are handled separately by the cross-field validation suite (Section 3.2). The five dimension scores $Q_d \in [0, 100]$ are:

Q_{loc} : Location (35%) Solution precision from the reported horizontal, depth, and time uncertainties (up to 40, 30 and 30 points of penalty, reached at 10 km, 10 km and 1 s). A missing uncertainty takes the penalty of the worst reportable value, so an unconstrained solution cannot score full marks.

Q_{net} : Network geometry (25%) Azimuthal gap, used-station count, and used-phase count (up to 50, 30 and 20 points of penalty). Azimuthal gap is among the principal predictors of horizontal location error (Bondár et al., 2004; Warren-Smith et al., 2025): a gap below 90° and a well-populated network (≥ 20 stations) each earn full marks on their respective (additive) terms, whereas a gap of 180° loses 45 points and few stations up to 30. Because $w_{\text{net}} = 0.25$, the gap term on its own can lower Q by at most 12.5 points; events with large gaps score low overall when, as usual, they also carry large location uncertainties and few stations. These cut-offs are CofC design heuristics; GeoNet location quality has been assessed against azimuthal-gap and station-coverage metrics of this kind (Warren-Smith et al., 2025).

Q_{sol} : Solution fit (15%) The RMS arrival-time residual (standard error) of the location: < 0.3 s earns full points, with progressively larger penalties above 0.5 and 1.0 s.

Q_{mag} : Magnitude (15%) Magnitude uncertainty and the number of stations contributing to the magnitude; small uncertainties and ≥ 10 amplitude measurements earn full points.

Q_{eval} : Evaluation (10%) Evaluation mode and status: manually reviewed or final solutions score highest; automatic or preliminary solutions are penalised, reflecting the additional confidence conferred by analyst review, and a solution its agency rejected earns nothing on this dimension.

Values outside a field’s physical range (for example a -999 missing-value sentinel) are treated as absent, and Q and every dimension score are clamped to $[0, 100]$. The score is stored with every event when it is written, including merged events, whose Q is that of the solution they publish.

Figure 6 summarises the five dimensions, their weights, and the grade mapping; the score Q maps to the letter grades in Table 3. Grades are displayed per event on the map and table views, and aggregated at catalogue level.

3.2 Cross-field validation

CofC applies a suite of physical-consistency checks at upload time and presents a summary before the user commits to saving:

- Magnitude-depth plausibility (e.g., $M > 8$ at depth < 5 km is extremely rare).
- Depth uncertainty exceeding twice the depth value.
- Magnitude uncertainty exceeding the magnitude for $|M| \geq 1$ (physically implausible).

Table 3: Quality grade thresholds. The A⁺–F scale provides an at-a-glance, ordinal event-level quality indicator derived from the continuous score Q of equation (1). The scoring dimensions follow established practice for catalogue-quality assessment (Woessner and Wiemer, 2005) and the network-geometry predictors of location accuracy (Bondár et al., 2004; Warren-Smith et al., 2025). Grades summarise the reported solution and metadata indicators, not event accuracy or fitness for a specific analysis, and the band boundaries are uncalibrated design choices (Section 10).

Grade	Score	Recommended use
A ⁺	95–100	Very strong solution indicators; verify suitability for the intended analysis.
A	85–94	Strong indicators; suitable for screening and candidate selection.
B ⁺	75–84	Acceptable; document limitations in methods.
B	65–74	Fair; notable gaps in uncertainty or provenance.
C	45–64	Poor; expert review recommended before quantitative use.
D	35–44	Minimal utility for quantitative analysis.
F	<35	Not recommended for quantitative analysis.

- Azimuthal gap $> 180^\circ$ (the recording stations span less than half the azimuths around the event), or a gap below the $360^\circ/N$ minimum that N stations can achieve.
- Used-station count exceeding the used-phase count.
- RMS travel-time residual < 0.001 s (suspiciously small, may indicate synthetic or processed data).
- Location-uncertainty anisotropy ratio > 10 (strongly elongated error ellipse suggesting poor azimuthal coverage or geometry).

Each check reports a severity level (*error*, *warning*, *info*) and a remediation suggestion. Checks of all three severities are surfaced to the user before saving; the checks are advisory and currently annotate rather than reject the affected event, so a record flagged *error* is still ingested and carries its flag.

4 Catalogue Merging

4.1 Duplicate detection

Duplicate detection uses an adaptive time-and-distance matching window. Two events from different source catalogues are considered duplicate candidates when both

$$|\Delta t| \leq \tau, \quad (2)$$

$$d \leq \delta, \quad (3)$$

where d is the great-circle distance computed from coordinates and the baseline windows τ and δ , which the user sets (the merge page starts at 60 s and 10 km; the user guide suggests 60 s and 50 km for a national catalogue), are widened automatically by fixed magnitude-dependent multipliers. In time the multiplier is 1.0/1.5/2.0/3.0 for $M < 4$, $4 \leq M < 5.5$, $5.5 \leq M < 7$, and $M \geq 7$ respectively; in distance it is 1.0/1.5/2.5/4.0 over the same bands, with an additional depth factor of 1.0/1.2/1.5 for depths < 100 , $100\text{--}300$, and > 300 km applied to the distance window. Association uses the great-circle *epicentral* distance; depth enters only through this widening factor, reflecting the poorer depth control of sparse networks. Magnitude is not part of the matching test; magnitude consistency is instead enforced as the per-group range gate described below. Near-antimeridian pairs use a great-circle distance and unit-vector coordinate averaging to avoid sign-wrapping errors in the Pacific region.

The baseline windows are user-configurable in the merge UI and are consistent in scale with the 10–20 km epicentral scatter documented for multi-network locations (Engdahl et al., 1998); configurable space–time association windows of this kind are also provided by catalogue-building toolkits (GEM Foundation, 2024). Association is one to one and best match first. Every pair of entries from *different* catalogues that fall inside each other’s widened windows is a candidate, ranked by its normalised separation $|\Delta t|/\tau + d/\delta$ (so time and distance count equally whatever the window sizes; magnitude agreement only breaks ties). Candidates are joined in increasing separation, so each entry is claimed by its closest counterpart in each other catalogue rather than by whichever earlier entry reached it first; a group never holds two entries from one catalogue (each catalogue is taken to be free of internal duplicates, and a merge request that lists the same catalogue twice is refused), and a group of three or more must have a member inside the window of every other member, so entries linked only through a chain are not fused. A group that fails the validity gates below is split into its largest self-consistent sub-groups (seeded from its earliest entry, candidates tried nearest first), and entries left unpaired may pair again in a further round. The result does not depend on the order of the input. The merge preview flags an *ambiguous association* when an entry’s best refused alternative was nearly as close as the pairing kept (within twice its separation plus a tenth of a window), since time and distance alone then barely separate the two choices.

Candidate groups that violate magnitude-scaled physical-validity gates are flagged as merge conflicts and are not merged. The magnitude gate compares the spread of a group’s magnitudes with a tolerance T of 0.5 ($\bar{M} < 4$), 0.8 ($\bar{M} 4\text{--}5.5$), 1.2 ($\bar{M} 5.5\text{--}7$), or 1.5 ($\bar{M} \geq 7$), $\bar{M}$ being the mean of the magnitudes as reported. Magnitudes are compared within each scale and, for a group mixing scales, on the common M_w scale against $\sqrt{T^{*2} + \sigma_{c,1}^2 + \sigma_{c,2}^2}$, $\sigma_{c,1}$ and $\sigma_{c,2}$ being the conversion uncertainties of the two extreme members; raw values on scales that cannot be converted are compared against T . The tolerance T^* of these single-scale

comparisons allows for the reported magnitude uncertainties σ_1 and σ_2 of the two extreme solutions (0 when not reported):

$$T^* = \min\left\{2T, \max\left[T, k\sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_{\text{ag}}^2}\right]\right\}, \quad (4)$$

with $k = 3$ (three-standard-deviation coverage) and $\sigma_{\text{ag}} = 0.2$, the scatter between two agencies' magnitudes of one earthquake that no reported uncertainty describes (the low end of the 0.2–0.3 typical of inter-agency M_L differences, so that no more scatter is credited than is observed). The cap of $2T$ (1.0 unit below $M4$) guards against bulletins that report the scatter of the station magnitudes rather than the error of their mean. The widening applies only when the association is uncontested (no refused alternative within the ambiguity margin above) and every pair of solutions in the group agrees closely in origin time and epicentre: within a fifth of its widened windows, $|\Delta t|/\tau + d/\delta \leq 0.2$ (for the 60 s and 50 km baseline, 12 s at one epicentre or 10 km at one origin time), and, where both solutions state them, within three combined standard errors of origin time and of epicentre. The bound is a fixed share of the windows, not a multiple of the reported errors, because it limits the chance that a second earthquake lies as close, which depends on the density of seismicity rather than on the location errors; it admits $0.2^3/3$, under 0.3%, of the space-time volume of the matching window. Otherwise $T^* = T$: where pairings are contested, as in a dense sequence, magnitude is the main evidence that tells neighbouring events apart. A depth fixed by the operator carries no depth information and is not compared; solved-for depths must agree within the larger of T_z and $3\sqrt{\sigma_{z,1}^2 + \sigma_{z,2}^2}$, $\sigma_{z,1}$ and $\sigma_{z,2}$ being the reported depth uncertainties of the shallowest and deepest solution and T_z being 30 km for shallow events (50 km from $M5$), with wider limits at intermediate and great depth. A group is also rejected if it has more than three members and spreads over more than 100 km ($M < 5$), 150 km ($M 5\text{--}6$), or 200 km ($M \geq 6$), or if more than 15 events fall in a single group. A group accepted only because of the reported uncertainties, or only because a fixed depth was left out, is not flagged, but the preview states why it was accepted.

4.2 Merge strategies

Once duplicate groups are identified, a conflict-resolution strategy is applied. CofC offers six strategies (Table 4).

Whatever the strategy, the published origin keeps its own metadata (agency, method, gap, counts, residual, evaluation status) and the published depth its own type and uncertainty: none of these is filled in from another entry, and a location or magnitude uncertainty is filled in only from an entry stating the same location, or the same magnitude value and type. The Average values strategy selects the magnitude rather than averaging it, and publishes it as reported, not converted. The preference depends on the event's size, taken as the median of the reported magnitudes on the M_w scale (M_L as M_w ; m_b and M_s by the relations of [Scordilis, 2006](#)). M_w comes first, with proxy M_w types ($M_w(m_B)$, M_{wp}) below true ones. Below $M6.2$, the M_L family (including GeoNet's summary magnitude M) comes before m_b , m_B and m_{bLg} , then M_s , then M_d ; from $M6.2$, M_s comes first, then m_B , M_L , m_b and m_{bLg} , and M_d . The switch is at 6.2 because there the M_s relation becomes one to one while M_L still reads M_w ,

Table 4: Merge strategies available in CofC. Whatever the strategy, every entry in a duplicate group is kept in the merged event’s lineage, with the entry whose solution was published marked (none, for an averaged or median event); no other entry’s origin or depth metadata is attached to the published solution (see text).

Strategy	Description
Quality-based (recommended)	Keeps the best-constrained entry under a merge-time solution score (station count, azimuthal gap, RMS residual, magnitude uncertainty and type, and evaluation status, on a fixed 100-point scale), comparing only the metrics <i>every</i> entry in the group states. If an entry states none of them, a network-authority ranking (a default table the administrator can edit) decides; remaining ties go to authority, then to the more complete core record. This merge-time score is distinct from the index Q (equation (1)); consolidating the two is planned.
Source priority	Keeps the entry of a chosen source. <i>Custom Order</i> : the user’s ranking of the selected catalogues decides, quality breaks ties, and unranked catalogues come last. <i>GeoNet (or GNS) > Others</i> : the GeoNet entry, recognised by its agency code or by the catalogue’s metadata or name, is kept; a group without one falls back to the network-authority ranking, with equally authoritative entries compared on the metrics they all state, as in the Quality-based strategy. The Quality-Based and Most Recent Solution options apply those rules.
Most recent solution	Keeps the solution the agency computed last (the origin’s QuakeML creation time, else the record’s creation or modification time). When not every entry states one, reviewed or final solutions win over preliminary ones, then quality; a rejected solution never wins. Upload time is not used.
Most complete	The entry with the greatest number of populated fields is kept, maximising metadata density in the merged product.
Average values	Only the epicentre is averaged: weighted by $1/\sigma^2$ when every entry states a horizontal uncertainty, with equal weights otherwise (no uncertainty is assumed for an entry that states none). Magnitude is selected by type, not averaged, and published as reported (see text). Depth, with its type and uncertainty, is taken from the best-constrained entry that solved for depth (a fixed depth only when none did); origin time is the earliest in the group; no single agency’s origin details (gap, counts, RMS) are carried onto the averaged epicentre. Best treated as exploratory, for closely consistent solutions.
Median values	As Average values, but the epicentre is the component-wise median of the reported latitudes and longitudes (longitudes unwrapped across the date line), every entry counting equally, and the origin time is the median of those reported (the mean, for two entries). Magnitude, depth and origin metadata are handled exactly as for Average values. One outlying entry moves neither coordinate; stated uncertainties are not used.

so the published value stays within about 0.1 of M_w on either side. Rejected magnitudes are skipped; ties go to the smaller uncertainty, then to the agency’s preferred magnitude.

Three refinements apply on top of any strategy. First, the depth, magnitude and focal mechanism of the merged event may each be resolved by a rule of their own rather than by the strategy: the depth from the best-constrained entry, or from the entry ranked first by quality, network authority or determination time, or from a named catalogue; the magnitude by the type preference above or from an entry chosen in the same ways; the focal mechanism from the united set (the default), from the published entry only, or from a named catalogue. The published depth and magnitude always carry their own metadata, and the entry that supplied each is marked in the lineage. Second, a group the preview flags (a failed validity gate, a split group, an ambiguous pairing, or an inconsistent magnitude or depth range) may be held for review instead of resolved: the strategy’s solution is written as provisional, the event is listed on the catalogue page with the reasons, and an editor either keeps that solution or publishes one contributing entry wholesale, which bumps the catalogue version. Third, two entries of one agency in a group (the agency of their catalogue, not merely the author of a copied origin) are compared on their agency event identifiers, reduced to a bare form: equal identifiers are vintages of one solution, of which only the most recently computed takes part, the older being kept in the lineage as superseded; differing identifiers are two earthquakes, and the group is split, each entry left alone being flagged with the reason; without comparable identifiers both entries take part as usual. The network-authority table that the Source priority fallback, the Quality-based tie-break and the authority rule consult (a global ranking with regional overrides for New Zealand and Japan by default) is editable by an administrator, and each merged event records which table was used.

The Quality-based strategy is recommended because it selects the best-constrained solution by an explicit, configurable rule rather than a subjective ranking of agencies, and it preserves all provenance information. Because several quality sub-scores increase with network density (Section 3), this strategy systematically favours solutions from denser networks; it is a reproducible, explicit preference, not a network-neutral one, and we recommend inspecting the per-strategy provenance split (Section 7) as a bias diagnostic. Focal mechanisms from all entries in a group are united, and the preferred one is chosen by source (Global CMT, then USGS/NEIC, GEOFON, GeoNet and INGV moment tensors, then other moment tensors, then first-motion solutions from at least 20 polarities, then automatic ones) and within a source by the quality of the solution.

Figure 3 shows the configuration and preview steps as the user encounters them. The matching windows and merge strategy are set explicitly, and the preview is a dry run of the merge they define, with a quality-control (QC) summary of it: the totals; for each source catalogue, how many of its entries were matched, how many appear only in it, and how often its solution was published; for each catalogue pair, the differences between matched solutions in origin time, epicentre, depth and magnitude (overall and by magnitude-type pair), as the median with a robust standard deviation (1.4826 times the median absolute deviation) and the 5–95% range, so that a systematic magnitude offset between two agencies shows directly; how many matched pairs lie near the edge of their windows, a measure of the merge’s sensitivity to the thresholds; and the flagged, kept-apart and held groups with their reasons. The groups to review are listed under *Flagged*, *Kept apart* (the entries of a group that failed the gates, published as separate events and shown together) and *Matched*, the

largest disagreement first (the greatest offset of an entry from the published solution, in units of its matching window); each group lists its entries with their Q and states which solution is published and why. The wizard’s one merge action, *Start Merge*, is enabled only once a preview exists for the current settings, and the merge stores the same summary with the merged catalogue, in the transaction that writes the catalogue; the summary is shown after the merge and on the merged catalogue’s page, and can be downloaded as JSON, its list of groups as CSV.

5 Statistical Analyses

5.1 Gutenberg-Richter b -value

The Gutenberg-Richter (GR) frequency-magnitude relation (Gutenberg and Richter, 1944) is

$$\log_{10} N(M) = a - bM, \quad (5)$$

where $N(M)$ is the cumulative number of earthquakes with magnitude $\geq M$, a is the $\log_{10}$ event-rate intercept, and $b \approx 1$ for tectonic seismicity (Gutenberg and Richter, 1944).

CofC estimates b using the maximum-likelihood estimator of Aki (1965); Utsu (1965):

$$\hat{b} = \frac{\log_{10} e}{\bar{M} - M_{\min} + \Delta M/2}, \quad (6)$$

where $\bar{M}$ is the sample mean magnitude, $M_{\min}$ is the lower magnitude cut-off (the estimated M_c , or an explicit cut-off; see Section 5.2), and ΔM is the step to which the magnitudes are *reported*, not the width of the plotted histogram bins (Bender, 1983): a magnitude rounded to 0.1 stands for an interval 0.1 wide, so the correction $\Delta M/2$ is 0.05 for magnitudes reported to 0.1, 0.005 for 0.01, and 0 for magnitudes reported to full precision (as GeoNet reports them). CofC detects the reporting step from the magnitudes themselves, counting a magnitude within 2^{-20} of a grid value as lying on it: the steps 0.1, 0.05, 0.01 and 0.001, and the coarse steps 0.2, 0.25 and 0.5 of historical and intensity-derived magnitudes when at least 95% of the sample lies on one of them. For a catalogue that mixes steps, such as a merged one, it applies the correction weighted by the share of the sample at each step, and it reports the step and the correction with each estimate. For the coarse steps the $\Delta M/2$ correction is itself approximate and leaves $\hat{b}$ 2–10% low at $b \approx 1$; the binned estimator of Bender (1983) is exact. Equation (6) is the uncorrected maximum-likelihood estimator and is biased high for small samples ($E[\hat{b}] = bN/(N-1)$; Marzocchi and Sandri, 2003), so at the minimum fitting count above an explicit cut-off ($N = 10$; Section 5.2) $\hat{b}$ is overestimated by $\sim 11\%$; we therefore recommend $N \gtrsim 200$ for reported b -values. The standard error on $\hat{b}$ is (Aki, 1965)

$$\sigma_b = \frac{\hat{b}}{\sqrt{N}}. \quad (7)$$

Equation (7) is the asymptotic counting-statistics error and *understates* the true uncertainty: it ignores the finite-sample variance of the observed magnitudes captured by the Shi and Bolt (1982) estimator and the magnitude-binning correction of Bender (1983), and it excludes

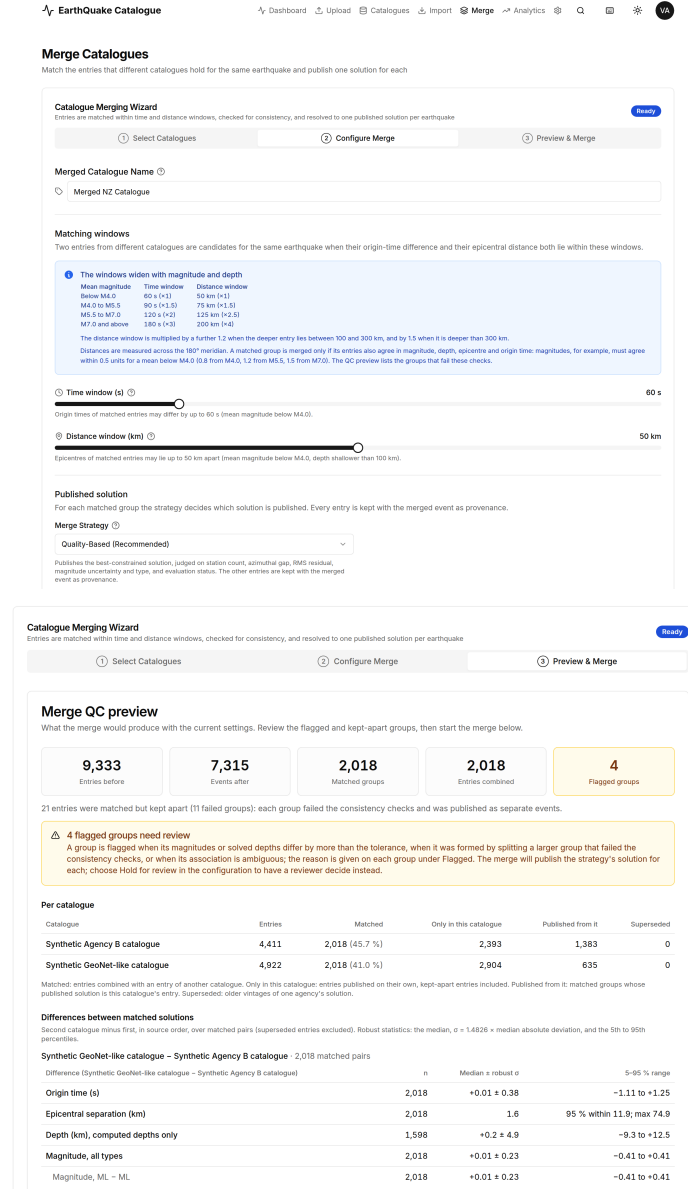


Figure 3: The merge workflow in the browser (the release described here). *Top*: the *Configure* step, with the matching windows (equations (2)–(3); here 60 s and 50 km), a table of the windows the adaptive multipliers give in each magnitude band (with the further widening for depth), and the strategy that selects the published solution (Table 4; here Quality-based). *Bottom*: the *Preview* step, a dry run of the merge with its QC summary, before anything is written. The totals read *Entries before* 9 333, *Events after* 7 315, *Matched groups* 2 018, *Entries combined* 2 018 and *Flagged groups* 4, and 21 entries of 11 groups that failed the gates are kept apart, published as separate events. The summary goes on to each catalogue’s matched entries (41.0% of the GeoNet-like and 45.7% of the Agency B entries) and the differences between matched solutions, and the groups to review follow under *Flagged* (4), *Kept apart* (11) and *Matched* (2 014), largest disagreement first. Data are a reduced-scale version of the synthetic pair of the worked example (Section 7), 4 922 GeoNet-like and 4 411 Agency B events built by the same generator with each true earthquake kept with probability 1/23.

the systematic sensitivity of $\hat{b}$ to the choice of M_c , to magnitude-scale heterogeneity, and to declustering, which usually dominate the total uncertainty (Section 8). We therefore treat the formal σ_b of Equation (7) as a lower bound and caution that realistic uncertainties are larger. CofC computes $\sigma_b = \hat{b}/\sqrt{N}$ and returns it with each Gutenberg-Richter estimate, displayed alongside the GR plot (cumulative and non-cumulative) and the fitted line.

5.2 Magnitude of completeness

The magnitude of completeness M_c is estimated by one of three methods, chosen in the M_c settings of the Analysis page. The default is the b -value stability method (MBS) of [Cao and Gao \(2002\)](#), in the form of [Woessner and Wiemer \(2005\)](#). Each bin edge M_i , in steps of 0.1, with at least 50 events at or above it in at least three populated bins is a candidate cut-off; $b(M_i)$ is estimated from the n events at or above it by equation (6) (with the reporting-step correction), with the uncertainty of [Shi and Bolt \(1982\)](#),

$$\delta b(M_i) = 2.3 b^2(M_i) \sqrt{\frac{\sum_{k=1}^n (M_k - \bar{M})^2}{n(n-1)}}, \quad (8)$$

and $\bar{b}(M_i)$ is the mean of b over the six cut-offs $M_i, M_i + 0.1, \dots, M_i + 0.5$, every one of them a candidate. M_c is the lowest M_i with

$$|\bar{b}(M_i) - b(M_i)| \leq \delta b(M_i), \quad (9)$$

so the test needs at least 50 events at or above $M_i + 0.5$, some 160 at or above M_i for $b = 1$. If no candidate passes, CofC reports the goodness-of-fit estimate below, or failing that the MAXC estimate, and says so. MBS is conservative by construction: it accepts a cut-off only where $\hat{b}$ has stopped changing, within its own uncertainty, over the next half magnitude unit. Because $\delta b \approx \hat{b}/\sqrt{n}$, about 0.005 for 40 000 events at $b = 1$, on a large catalogue it also rejects cut-offs at which $\hat{b}$ still drifts by only a few hundredths, whether from residual incompleteness, a mixture of populations with different b -values, or an excess of large events over the GR law, and M_c rises accordingly: on the worked example's catalogues of about 10^5 events it lies 0.7–0.9 units above the goodness-of-fit estimate (Section 7). Whichever method is selected, the M_c tab therefore plots $b(M_i) \pm \delta b(M_i)$ and $\bar{b}(M_i)$ against the cut-off, with M_c marked, so that a cut-off raised by a drift the analyst judges immaterial can be recognised and replaced by an explicit one.

The first alternative is the Goodness-of-Fit Test (GFT) of [Wiemer and Wyss \(2000\)](#), in the form of [Woessner and Wiemer \(2005\)](#). Each bin edge M_i with at least 10 events at or above it in at least three populated bins is a candidate cut-off; b is estimated above it by equation (6) (with the same reporting-step correction), the predicted cumulative counts $S_j = N(\geq M_i) 10^{-b(M_j - M_i)}$ are compared with the observed counts $B_j = N(\geq M_j)$ at every bin edge $M_j \geq M_i$ up to the bin holding the largest magnitude, and the fit is $R = 100 - 100 \sum_j |B_j - S_j| / \sum_j B_j$. M_c is the lowest candidate with $R \geq 95\%$, else the lowest with $R \geq 90\%$; if no candidate reaches 90%, CofC reports the MAXC estimate and says so. The Analysis page plots R against the candidate cut-off, with the chosen M_c marked. The second alternative is the Maximum Curvature (MAXC) method ([Wiemer and Wyss, 2000](#)):

$$M_c^{\text{MAXC}} = \arg \max_M N_{\text{non-cum}}(M) + c, \quad (10)$$

i.e. the centre of the 0.1-wide magnitude bin containing the most events in the non-cumulative frequency-magnitude distribution (the lowest such bin on a tie), plus a correction c . As in [Wiemer and Wyss \(2000\)](#), each magnitude is binned to its nearest multiple of 0.1, so the bin labelled M holds $[M - 0.05, M + 0.05)$ and the M_c tab plots the same bins; for magnitudes reported to 0.1 this is the reported value, whereas labelling the bins by their lower edges would bias the estimate for full-precision magnitudes low by half a bin on average. MAXC is fast and makes no assumption about the GR model, but tends to underestimate M_c by ~ 0.1 – 0.2 magnitude units for typical networks ([Woessner and Wiemer, 2005](#)); CofC therefore adds $c = 0.2$ by default (also when MBS or the GFT falls back to MAXC), which the user can set from 0 to 0.5 in steps of 0.1, and every result states the correction it used.

The formal uncertainty on M_c is at least ± 0.1 (one bin width), but [Woessner and Wiemer \(2005\)](#) showed empirically that the true uncertainty is typically 0.2–0.3 magnitude units, for MAXC because the non-cumulative FMD peak is broad and sensitive to sample fluctuations. CofC displays M_c with the bin width as a lower bound on its uncertainty and recommends visual verification against the FMD and b -value stability plots. All three methods estimate a single M_c for the events under analysis; where completeness varies in space or time, as in most merged catalogues, the MAXC and goodness-of-fit values in particular can lie well below the completeness of parts of the catalogue (Section 7). The Entire Magnitude Range method ([Woessner and Wiemer, 2005](#)) remains external post-processing.

A commonly adopted minimum for a reliable b -value estimate is $N \approx 200$ events above M_c (cf. [Marzocchi and Sandri 2003](#)); below this the formal $\sigma_b = \hat{b}/\sqrt{N}$ exceeds ≈ 0.07 (for $\hat{b} \approx 1$), comparable to the biases discussed in Section 8, and the confidence interval on b becomes too wide for quantitative hazard use. CofC enforces hard floors below which an estimate is withheld rather than reported: estimating M_c requires at least 50 events; a Gutenberg-Richter fit requires at least 10 events at or above its cut-off, spread over at least three populated magnitude bins (and, when the fit estimates its own M_c , the 50 events above); and the b -value of an individual aftershock sequence is reported only for sequences of at least 50 events, above the sequence’s own M_c (always by b -value stability, with its fallbacks, whatever method the M_c settings select), since short-term incompleteness early in a sequence raises its M_c above the background network’s. The $N \approx 200$ figure is a guideline the analyst should apply when interpreting results.

5.3 Declustering

Aftershock sequences violate the Poissonian independence assumption underlying GR analysis. CofC implements the Gardner-Knopoff window method ([Gardner and Knopoff, 1974](#)) with the commonly tabulated magnitude-dependent windows (as compiled by [van Stiphout et al. 2012](#)). For each event the space-time interaction zone is:

$$L(M) = 10^{0.1238M+0.983} \text{ km}, \quad (11)$$

$$T(M) = \begin{cases} 10^{0.032M+2.7389} \text{ days}, & M \geq 6.5, \\ 10^{0.5409M-0.547} \text{ days}, & M < 6.5 \end{cases}, \quad (12)$$

the two-branch time window $T(M)$ following the larger-magnitude relation for $M \geq 6.5$ and the small-magnitude relation otherwise. Events are visited in order of decreasing magnitude,

and every not-yet-assigned event within $L(M)$ and $T(M)$ of the visited event is assigned to its cluster; each visited event is itself reserved, so a later, smaller event cannot absorb a larger one. CofC applies $T(M)$ *forward* in time from the mainshock, as in the original [Gardner and Knopoff \(1974\)](#) procedure and as the OpenQuake `hmtk` reference implementation does with its foreshock-window proportion (`fs_time_prop`) set to zero; foreshocks are therefore not removed. (A symmetric window that also reaches back $T(M)$ before each mainshock is a common variant; on the synthetic catalogue of Section 7 it removes a further 970 events above the analysis cut-off and lowers the declustered $\hat{b}$ by 0.06, about four formal standard errors, so the choice is not cosmetic and is stated here.) The Reasenber (1985) algorithm, which uses an interaction zone based on the Omori–Utsu law ([Omori, 1894](#); [Utsu, 1961](#)) and adapts to the local seismicity rate, is implemented in the analysis library as an alternative, though the browser worker currently exposes only the Gardner-Knopoff method.

Declustering returns a mainshock/aftershock partition for the events under analysis; an event whose origin time cannot be parsed falls in no window and is left independent, and the Temporal tab reports how many such events there are. Exports can carry it: with the export option for Gardner-Knopoff tags, each exported event gains its cluster (identified by its mainshock) and a mainshock flag, and the export records the algorithm, the windows used and the counts; writing the partition back as a separate catalogue within the platform is not implemented. Events preceding a larger mainshock are retained; the declustered set is the mainshock population in the usual window-method sense ([van Stiphout et al., 2012](#)).

5.4 Cumulative seismic moment

Cumulative seismic moment is computed as

$$M_0 = 10^{1.5M_w+9.1} \text{ N m}, \quad (13)$$

using the Hanks-Kanamori relation ([Hanks and Kanamori, 1979](#); [Kanamori, 1977](#)) in SI units (equivalently $M_w = \frac{2}{3} \log_{10} M_0 - 6.07$ for M_0 in N m). The sum applies Equation (13) to M_w as reported and, assuming $M_L \approx M_w$, to M_L , to GeoNet’s summary magnitude M and to untyped magnitudes; m_b , m_B , M_s , M_d and other stated types are excluded, and the Analysis page states how many events were treated each way. (The library’s empirical conversions, after [Scordilis \(2006\)](#) for m_b and M_s , serve the merge’s magnitude-consistency gate and its estimate of an event’s size for the magnitude-type preference; automatic homogenisation of mixed magnitude types within the analysis pipeline remains future work, Section 8.) Because cumulative moment is dominated by the largest events, magnitude saturation (Section 8) in any saturated input magnitudes propagates strongly to $\sum M_0$; $\sum M_0$ is therefore only physically meaningful when the input magnitudes are moment magnitudes or close proxies for them, and totals that rely on the $M_L \approx M_w$ assumption should be treated as indicative. The Temporal tab plots the cumulative release of seismic moment, or of radiated energy ($\log_{10} E = 1.5M + 4.8$, in joules; for M_w this is $E = M_0/(2 \times 10^4)$, [Kanamori, 1977](#)), against time for the same eligible events.

5.5 Temporal pattern analysis

Beyond the aggregate GR analysis, CofC provides two temporal diagnostics for detecting non-stationarity in the catalogue.

Seismicity rate time series. Events at or above a magnitude threshold are counted per time bin (one UTC day, ISO week or calendar month, chosen by the user, or by default daily for spans up to a year and weekly beyond) and plotted as a rate time series. The threshold is the magnitude filter’s lower bound when one is set, and otherwise the M_c estimated under the current M_c settings; with fewer than 50 events, too few to estimate M_c , every event is counted and the page says that the rate then also follows changes in detection. Partial first and last bins are scaled or marked as described in Section 2.4, and events whose origin time cannot be parsed are left out of the series and counted. Sudden rate changes, which often correspond to network upgrades, station outages, or changes in processing procedures, are immediately visible and can be used to identify breakpoints for time-varying M_c analysis (Mignan and Woessner, 2012).

Interevent-time distribution (specified, not yet implemented). The distribution of times between successive events above M_c is to be plotted against the exponential distribution expected for a Poissonian process. Systematic departures, most commonly the excess of short interevent times produced by aftershock clusters, provide a visual diagnostic of whether the catalogue has been adequately declustered before analyses that assume Poissonian seismicity. The exponential reference assumes a stationary rate; for strongly non-stationary catalogues the time-rescaling (residual) approach of Ogata (1988) should be applied before interpreting departures as clustering.

6 Export Formats

Processed or merged catalogues can be exported in:

- **CSV:** a flat table, one row per event, with the stored fields and per-event lineage columns (source-catalogue IDs, merge strategy and parameters, the selected source entry and its catalogue, quality score and grade, review status, the agency’s own event type, and the catalogue version), suitable for ZMAP (Wiemer, 2001), ObsPy (Beyreuther et al., 2010), and R/Python workflows; an optional commented prologue carries the catalogue metadata.
- **QuakeML 1.2 BED** (Schorlemmer et al., 2011): re-serialisation of the stored origins, magnitudes, and extended fields (a merged event’s published solution is either the contributing agency’s own origin or a new origin identified as merged), with each event’s lineage as an event-level `comment`, catalogue-level merge metadata as `comment` elements, and the catalogue version in the creation information; the output validates against the QuakeML-BED 1.2 schema.
- **JSON and GeoJSON** (Butler et al., 2016): the same per-event fields and lineage as properties, with the merge configuration and source catalogues in the document head; GeoJSON suits GIS software and web mapping.
- **KML:** Open Geospatial Consortium Keyhole Markup Language, for Google Earth and similar viewers, with magnitude-scaled symbols.

Exports may be taken for a whole catalogue or for the events matching the catalogue page’s filters (magnitude, depth and time ranges, a bounding box that may cross the antimeridian, event and magnitude types, evaluation status and mode, azimuthal gap, station and phase counts, RMS residual, per-field uncertainty limits, and a minimum quality score). Every export records the catalogue’s identifier and version (with the time that version was released), the export time (UTC), the filter and any declustering applied (Gardner-Knopoff tags are an option; Section 5.3), and a SHA-256 checksum over the exported event rows that is identical in every format for the same selection; the version also appears in the file name. A study can therefore cite the version it used and check a copy against it; because the platform does not keep superseded versions, the export itself, or an archive deposit of it, is the record of that data state (Section 2).

7 Worked Example: New Zealand Seismicity

To illustrate the CofC workflow and quantify the statistical biases described in Section 8, we construct a *synthetic* catalogue pair representative of two agencies monitoring New Zealand seismicity and process it with the platform’s own engine. The construction is a seeded model, not a fit to any real dataset; its parameters are listed in the electronic supplement (Table S1). One population of true earthquakes spans January 2020–December 2024: an independent background with a Gutenberg-Richter $b = 1.00$ spread over seven New Zealand source zones, at about 3 200 events of $M \geq 2.3$ per year, plus aftershock sequences with $b = 1.30$ injected on the background events of $M \geq 4.5$ (Omori-law timing, Omori, 1894; Utsu, 1961), sized so that 23% of the true events of $M \geq 2.0$ are aftershocks. Two synthetic station networks record them: a GeoNet-like national network of 166 stations at ~ 40 km spacing on land, and an Agency B regional network of 67 stations at ~ 30 km spacing in the central and eastern North Island (Figure 4a). An agency catalogues every earthquake that four or more of its stations record within a magnitude-dependent distance. Each entry’s azimuthal gap, station count and nearest-station distance follow from its network’s geometry; its location, depth, origin-time and magnitude uncertainties follow stated scalings with those quantities (growing with the gap and with the distance outside the network, falling with the number of stations); and its reported values are the true ones perturbed by errors of that size. Both agencies report analyst-reviewed M_L , the GeoNet-like agency to full precision and Agency B to 0.1. The resulting catalogues hold 113 427 (GeoNet-like) and 101 806 (Agency B) events, of the order of the GeoNet national catalogue over the same period. Every number below is computed from these catalogues by the platform’s own modules, called as the platform calls them (Data and Resources); the generator’s ground truth is used only to score the results.

7.1 Step 1: Import and quality scoring

The catalogues are loaded with the fields an uploaded QuakeML file stores, and every event is scored with equation (1) on import. The median is $Q = 69$ (grade B) for both catalogues. The azimuthal gap (Figure 5a) exceeds 180° for 25% of the GeoNet-like and 29% of the Agency B events: the offshore events and, for Agency B, those outside its regional network. Equation (1) ranks these events lowest (Figure 5b): the median Q falls from 87 for gaps below 90° to 53–55

between 180° and 270° and 35–36 above. At most 12.5 of those points come from the gap term itself; the rest come from the larger location uncertainties and smaller station and phase counts that the high-gap entries carry, a correlation built into the synthetic uncertainty model and one that real network locations also show (Bondár et al., 2004). The platform’s MAXC estimates of completeness are $M_c = 1.8$ for the GeoNet-like catalogue and 1.4 for Agency B, whose denser network detects smaller events in its region (the goodness-of-fit test gives 1.9 for both). Its default, b -value-stability estimates are $M_c = 2.7$ and 2.6: with 10^5 events, δb at the lower cut-offs is a few thousandths, so the test rejects every cut-off at which $\hat{b}$ still differs by more than that from its mean over the next half unit (Section 5.2; Step 4).

7.2 Step 2: Merge

Merging with the Quality-based strategy and baseline windows of 60 s and 50 km (equations (2) and (3)), the matcher forms 47 384 duplicate groups, and the $113\,427 + 101\,806 = 215\,233$ records merge into 167 849 events (Figure 4b): 66 043 reported by the GeoNet-like agency only, 54 422 by Agency B only (mostly small events inside its denser network), and 47 384 by both. Because the generator knows which entries describe the same earthquake, the association can be scored exactly. Of the 47 109 earthquakes both agencies report, 46 640 (99%) are paired correctly, and 98.4% of the groups join two entries of one earthquake. Of the 744 wrong pairings, 741 lie within aftershock sequences, where several events fall inside one window, and none is among the 2 029 groups that pass the magnitude gate only within the tolerance widened by the reported uncertainties (equation (4)). Of the 469 pairs left apart, the validity gates reject 285 whose two magnitudes differ by more than the gate allows (0.5 units below M_4 , at most 1.0 where the tolerance widens) and 16 whose solved-for depths differ by more than it allows (30 km for shallow events, or three combined standard errors); 141 lie outside the matching window (distant events Agency B locates poorly), and in 27 each entry has a closer counterpart. These earthquakes stay in the merged catalogue twice. The Quality-based strategy keeps Agency B’s solution for 32 012 of the duplicate groups (68%), those inside its denser network, and the GeoNet-like solution for the other 15 372; every contributing entry is kept in the merged event’s lineage, and each exported event carries its source catalogue IDs, the merge strategy, the selected entry and its quality score. These recall and precision figures reflect the synthetic location, timing and magnitude scatter; they are a check of the association logic, not a benchmark for real catalogue pairs, whose errors are non-Gaussian and whose clocks and magnitude scales differ (Section 10).

7.3 Step 3: Quality filtering

A quality threshold of $Q \geq 70$ (upper grade B and above) retains 86 028 of the 167 849 merged events (51.3%). It removes 94.3% of the 37 678 merged events with gap $> 180^\circ$, which make up 43% of the removed set, so the filter does remove most of the poorly constrained offshore population. It is not neutral in magnitude, however. Because Q rises with the number of recording stations and falls with the location and magnitude uncertainties, all of which depend on magnitude, the filter keeps 59% of the events of $M_{1.5-2.0}$ but 86% of those of $M_{3.5-4.0}$, and so changes the shape of the frequency-magnitude distribution (Figure 7a).

7.4 Step 4: Completeness and b -value

On the quality-filtered catalogue the platform’s MAXC estimate is $M_c = 1.8$ (MAXC with the +0.2 correction), above which the maximum-likelihood fit gives $\hat{b} = 0.837 \pm 0.004$ ($N = 36\,824$); the goodness-of-fit test gives $M_c = 1.9$ at the 95% level and $\hat{b} = 0.863 \pm 0.005$. Both lie far below the planted values, because completeness varies across the merged catalogue: Agency B’s network detects small events in the central North Island that the GeoNet-like network misses, and both networks miss small offshore and deep events. A single global M_c from either method is set by the best-covered part of the catalogue, and the catalogue above it is incomplete elsewhere. Stepping the magnitude cut-off shows $\hat{b}$ rising until $M2.7$ and stable above it (Figure 7c), and the platform’s default estimate finds this: b -value stability (Section 5.2) gives $M_c = 2.7$ for the merged catalogue, which we adopt as the cut-off for the comparisons below, and 2.8 for the quality-filtered catalogue, whose $\hat{b}$ still rises by about 0.01 per 0.1 unit up to $M3.0$. Above $M2.7$ the merged catalogue gives $\hat{b} = 1.061 \pm 0.012$ ($N = 8\,304$), a mixture of the independent background and the injected aftershocks. Applied to the background events alone, the same routine returns $\hat{b} = 1.011 \pm 0.012$ against the planted 1.00, and to the injected aftershocks alone 1.35 ± 0.04 against the planted 1.30: the estimator is sound once the sample is complete. The quality-filtered catalogue above $M2.7$, in which 15% of the events are injected aftershocks, gives $\hat{b} = 0.998 \pm 0.012$ ($N = 6\,446$; Figure 7b), 0.06 below the unfiltered value at the same cut-off, and its background events alone give 0.955 ± 0.013 : the magnitude selectivity of the filter persists above the completeness magnitude. These are checks of the platform’s behaviour on a controlled input, not measurements of New Zealand seismicity.

7.5 Step 5: Declustering

Gardner-Knopoff declustering (Section 5.3) of the 6 446 quality-filtered events above $M2.7$ removes 2 525 of them (39%, against the 15% injected as aftershocks). The windows recover 96.5% of the injected aftershocks but also sweep up 1 606 independent events that fall inside the space-time window of a larger event, and these are preferentially small (their own $\hat{b}$ is 1.19), because every cluster head survives while small events are the ones absorbed. The declustered estimate, $\hat{b} = 0.883 \pm 0.014$, therefore lies 0.11 below the undeclustered value and 0.12 below the planted background $b = 1.00$: window declustering shifts b downwards, as Mizrahi et al. (2021) found for real catalogues, and the shift here is not the removal of an aftershock excess alone. Because the synthetic aftershocks follow their mainshocks, the forward window matches this construction; the symmetric window convention removes a further 968 events and gives $\hat{b} = 0.824 \pm 0.015$, 0.06 (about four formal standard errors) lower still. Figure 8 summarises the event-count flow through these steps.

8 Guidance on Avoiding Statistical Biases

This section gives the practical mitigation for the biases that arise when working with merged multi-agency catalogues. The underlying mechanisms are described in Section 1.4; here we recap each only briefly and concentrate on the corrective steps and how CofC supports them. These biases are not unique to CofC; they affect any merged catalogue analysis.

8.1 Magnitude completeness bias

Problem. Fitting b below M_c underestimates it, because the incomplete sample is deficient at small magnitudes (Section 1.4). Completeness also varies in space: a merged catalogue can be complete to low magnitudes where one agency’s network is dense and only to much higher ones offshore or at depth.

Guidance.

1. Always estimate M_c before fitting b , and verify the estimate by eye on the FMD and b -value stability plots rather than adopting it unexamined. In the synthetic example (Section 7) the MAXC and goodness-of-fit estimates ($M1.8$ and $M1.9$) lie well below the $M2.7$ from which $\hat{b}$ is stable, which the default b -value-stability estimate finds. That estimate errs the other way: on a large catalogue, or one with an excess of large events over the GR law, it can rise above a cut-off the analyst would accept (Section 5.2). Where the stability plot shows only a drift the analyst judges immaterial, set an explicit cut-off (the magnitude filter’s lower bound) and report it.
2. Set the lower magnitude cut-off slightly above M_c (e.g. $M_c + 0.1$) as a practical conservatism against partial completeness in the transition zone.
3. After merging catalogues with different M_c values, use at least the *higher* of the two as the common cut-off. This is necessary but not sufficient: in the synthetic example the higher of the two catalogues’ MAXC estimates ($M1.8$, or $M1.9$ by the goodness-of-fit test) is still below the merged catalogue’s completeness, because completeness also varies within each network; the higher of their b -value-stability estimates ($M2.7$) reaches it.

8.2 Magnitude heterogeneity bias

Problem. Mixing magnitude scales artificially broadens the FMD and can shift the apparent b -value, in part because different scales saturate at different magnitudes (Section 1.4).

Guidance.

1. Inspect the magnitude-type distribution before fitting: the Analysis page tabulates the magnitude types of the events under analysis, warns when the Gutenberg-Richter or M_c sample mixes scale families, and can restrict the analysis to chosen types.
2. Convert to a single preferred scale (ideally Mw) using empirical regressions (Das et al., 2011; Hanks and Kanamori, 1979), preferably orthogonal rather than ordinary least-squares fits, to avoid conversion bias (Castellaro et al., 2006; Lolli and Gasperini, 2012), where possible.
3. If conversion is not possible, split the catalogue by magnitude type and fit each subset separately.

8.3 Network coverage and azimuthal gap bias

Problem. Poorly located events from sparse-network regions can bias statistical distributions if they concentrate at particular magnitudes or depths (Section 1.4). Removing them is not

free: because location quality improves with magnitude, a quality filter removes small events preferentially.

Guidance.

1. Filter on quality score (e.g., $Q \geq 45$, grade C or better) or azimuthal gap (e.g., gap $\leq 170^\circ$, in the page’s 10° steps) before location-sensitive analyses such as spatial mapping. Records the reporting agency flagged as duplicate, not existing, or not locatable are excluded from the Analysis page by default.
2. Before fitting b to a quality-filtered catalogue, compare the frequency-magnitude distributions or b -values with and without the filter at the same cut-off: in the synthetic example a $Q \geq 70$ filter keeps 59% of the events of $M1.5$ – 2.0 but 86% of those of $M3.5$ – 4.0 , and lowers $\hat{b}$ by 0.06 even above the completeness magnitude.
3. Check the spatial distribution of excluded events; geographic concentration may indicate a bias rather than a data problem.
4. Use the CofC map coloured by quality grade or by azimuthal gap to identify network-coverage artefacts.

8.4 Aftershock clustering and GR analysis

Problem. GR analysis assumes independent, Poissonian arrivals. Aftershock sequences cluster in space, time, and magnitude, violating this assumption and shifting the estimated b . Declustering shifts it too: window methods remove small independent events more readily than large ones (Mizrahi et al., 2021). In the synthetic example of Section 7, where the clustered population was assigned a higher b by construction, the quality-filtered catalogue gives $\hat{b} = 1.00$ above $M2.7$ before declustering and 0.88 after, below the planted background value of 1.00, because the windows also remove 1 606 small independent events. The declustered value also depends on the window convention: a symmetric window gives 0.82 on the same catalogue (Section 5.3), a difference of about four formal σ_b , so the convention must be reported alongside any declustered b .

Guidance.

1. Decluster before analyses that assume Poissonian seismicity. The Gardner-Knopoff method (Gardner and Knopoff, 1974) implemented in CofC is the standard choice for tectonic seismicity; Reasenber (1985) is preferable for swarm-dominated sequences.
2. For b -values, compare the original and the declustered catalogue: export the catalogue with the Gardner-Knopoff tags and fit both, treating the difference as a measure of the preprocessing uncertainty rather than as a correction.
3. Report the declustering algorithm and parameters in publications so results are reproducible.

8.5 Duplicate double-counting

Problem. Merging without duplicate removal counts each shared earthquake once per entry, inflating $N(M)$ wherever the catalogues overlap, raising the estimated a -value, and distorting rate estimates. In the synthetic example, concatenating the two catalogues instead of merging them would raise the number of events above $M2.7$ from 8 304 to 13 312 (60% more) and the a -value from 6.78 to 6.87; because the shared events are the larger ones that both networks record, it would also lower $\hat{b}$, from 1.06 to 1.02. The Analysis page therefore withholds the b -value, M_c , rates and moment totals for a set pooled from several catalogues.

Guidance.

1. Always merge catalogues before statistical analysis; never concatenate raw files.
2. Inspect the merge’s QC summary (matched, flagged and kept-apart groups, the differences between matched solutions, the pairs near the edge of their windows, strategy applied), which is kept with the merged catalogue, and satisfy yourself that the matching thresholds are physically appropriate for your region and magnitude range.
3. For teleseismic analysis, increase the distance threshold (e.g., to 100 km) to account for larger location scatter at global distances.

8.6 Temporal non-stationarity

Problem. Network upgrades, station outages, or changes in analysis procedures alter M_c over time. A single b fitted over a long catalogue period conflates eras with different completeness.

Guidance.

1. Use the CofC Temporal tab to identify sudden jumps in the rate of events above M_c or in the magnitude–time distribution, likely indicators of network changes.
2. Split the catalogue at breakpoints and estimate b for each period separately.
3. The [Mignan and Woessner \(2012\)](#) review provides detailed methods for time-varying M_c analysis.

9 Software Availability

CofC is released under the MIT licence (a LICENSE file is included in the repository). The source code, issue tracker, and documentation are available at:

<https://github.com/KennyGraham1/catalogofcatalogs>

Full documentation (user guide, developer guide, API reference, deployment guide) is hosted at:

<https://catalogofcatalogs.readthedocs.io/en/latest/>

The repository also includes Docker Compose recipes for single-command deployment and a continuous-integration pipeline that runs linting, type-checking, and the Jest test suite (covering the parsing, merging, quality-scoring, and seismological-analysis modules and the worked-example engine) on Node.js 22 and 24. For citation of a fixed software state, the release described here will be tagged in the repository and deposited in Zenodo to mint a persistent DOI; a `CITATION.cff` recording that identifier will be added to the repository, and the DOI will be cited in this section, upon acceptance.

System requirements

- Node.js 22 (22.12 or later) or 24
- MongoDB $\geq 6.x$ (local or Atlas cloud)
- Modern browser (Chrome, Firefox, Edge, Safari)

Dependencies

Key third-party dependencies (all MIT or compatible licences): Next.js ([Vercel Inc., 2024](#)), MongoDB Node.js driver, Leaflet ([Agafonkin, 2024](#)), Apache ECharts ([The Apache Software Foundation, 2024](#)), Tailwind CSS, shadcn/ui, NextAuth.js (authentication), and sax (streaming QuakeML parsing).

10 Discussion and Conclusions

10.1 Relation to existing approaches

CofC addresses a gap that is simultaneously methodological and infrastructural. The methodological gap is well documented in the CORSSA literature ([Naylor et al., 2010](#); [Mignan and Woessner, 2012](#); [van Stiphout et al., 2012](#)): the biases described in Section 8 are known, quantifiable, and avoidable, yet they continue to appear in published seismicity studies because the corrective steps (enforcing M_c , homogenising magnitude scales, declustering, and quality filtering) are each individually inconvenient to apply and are rarely performed together in a reproducible workflow. The infrastructural gap, that no single production-ready tool spans multi-source ingest, merging, quality scoring, and the statistical workflow, is set out against existing tools in Section 1.5 and Table 1. What CofC adds beyond that comparison is to close the loop end to end, from multi-source ingest through duplicate removal, quality scoring, declustering, and GR fitting, within a single persistent web interface that preserves provenance at every step and requires no local installation.

10.2 Quality scoring and event selection

The quality-scoring system (equation (1)) is designed to be interpretable: each sub-score corresponds to a well-understood determinant of solution quality, grounded in the seismological literature. Azimuthal gap is among the principal network-geometry predictors of horizontal

location error (Bondár et al., 2004); used-station and used-phase counts further constrain the solution; location and magnitude uncertainties are direct quantitative measures of solution precision. By making these criteria explicit and combining them into a single 0–100 index, CofC provides a reproducible basis for quality-based event selection that does not depend on subjective judgement of agency provenance. Because the sub-scores scale with network density, however, this ranking favours densely recorded events and is not network-neutral (Section 3); and because they scale with the number of recording stations, it also favours larger events, so a threshold on Q is a magnitude-dependent selection (Section 7).

An alternative quality-filtering approach common in the literature is to apply hard thresholds on individual parameters (e.g., retain only events with $\text{gap} < 180^\circ$ and $\sigma_M < 0.3$). Such thresholds are easier to report in a methods section but are less flexible: a threshold approach discards events entirely, whereas the quality score allows users to weight the trade-off between sample size and data quality continuously. CofC exposes both approaches: the catalogue page and the filter API select events by a minimum quality score or by per-field limits (azimuthal gap, station and phase counts, RMS residual, and the location, depth, time and magnitude uncertainties), and the Analysis page filters by Q and by azimuthal gap.

10.3 Towards reproducible seismicity analysis

A persistent challenge in observational seismology is that two researchers applying the same analysis method to nominally the same dataset can obtain different results, because their preprocessing choices (format conversion, duplicate removal, M_c cut-off, declustering algorithm and parameters, quality filters) differ in ways that are rarely fully reported (Schorlemmer et al., 2007). CofC addresses this at the infrastructure level: every catalogue stored in the platform carries a provenance record (source catalogue IDs and contributing entries per event, merge strategy and parameters, quality score) and a version. Every export includes this record together with the catalogue version, a checksum of the exported rows, the filter applied and, when requested, the declustering algorithm, its parameters and per-event tags, and the QuakeML export format (Schorlemmer et al., 2011) provides a standard container for sharing it. The platform thus encourages a workflow in which the preprocessing choices are not buried in bespoke scripts but are captured as persistent, shareable catalogue objects.

10.4 Limitations and future work

M_c estimation. The MAXC method (Wiemer and Wyss, 2000) implemented in CofC is fast and requires no assumptions about the GR model, but Woessner and Wiemer (2005) showed that it systematically underestimates M_c by ~ 0.1 – 0.2 magnitude units in typical network conditions; the $+0.2$ correction mitigates this but does not eliminate it. On a merged catalogue whose completeness varies in space, the worked example shows MAXC and the goodness-of-fit test both falling well below the magnitude from which b is stable (Section 7). The default b -value-stability estimate (Cao and Gao, 2002; Woessner and Wiemer, 2005) reaches that magnitude, but it errs in the other direction: its tolerance δb shrinks as $1/\sqrt{N}$, so on large catalogues it rejects cut-offs at which $\hat{b}$ still drifts by only a few hundredths, and an excess of large events over the GR law raises it further. Its M_c is therefore conservative and should be checked against the stability plot. All three methods estimate one M_c for

the events under analysis. The Entire-Magnitude-Range method (Woessner and Wiemer, 2005), spatially varying M_c (Schorlemmer and Woessner, 2008), and, for time-varying M_c , the bootstrap approach described by Mignan and Woessner (2012) are planned.

Spatial analysis. CofC currently provides only global (single value) b -value estimates. Spatial mapping of b (Wiemer and Wyss, 2000; Schorlemmer et al., 2005), of M_c (Schorlemmer and Woessner, 2008), and of seismicity rates on a grid or along a fault structure requires server-side spatial indexing and is targeted for a future release. Similarly, ETAS parameter fitting (Ogata, 1988) and forecast evaluation using the CSEP framework and its consistency tests (Schorlemmer et al., 2007; Zechar and Jordan, 2010) are not yet implemented.

Scale. Catalogues larger than $\sim 10^6$ events require server-side pre-aggregation (e.g., hexagonal binning for the map view, streaming computation for GR fits) for smooth browser rendering. MongoDB’s aggregation pipeline and server-side sampling, already used for the API pagination layer, will be extended to support these workflows.

Magnitude homogenisation. CofC displays the magnitude type of each event, tabulates the types under analysis, and warns when a fit mixes scales, but it does not yet apply empirical inter-scale conversion automatically. Given the magnitude bias this can introduce (Marzocchi and Sandri, 2003; Tinti and Mulargia, 1987), automated conversion to a unified scale (ideally M_w) using orthogonal regressions such as those of Das et al. (2011) and Lolli and Gasperini (2012) is a high-priority addition; database-scale homogenisation workflows such as that of Weatherill et al. (2016) provide a template.

10.5 Conclusions

We have presented CofC, a browser-based platform for earthquake catalogue management, merging, quality scoring, and seismological analysis. The core contribution is infrastructural: a single persistent interface that integrates format ingestion, duplicate detection, transparent quality scoring, and provenance-preserving export, the steps that are individually inconvenient yet collectively determine the validity of downstream statistical analysis. Using a fully synthetic two-catalogue New Zealand example processed end to end by the platform’s own engine, we illustrated the workflow and quantified what it exposes: the matcher pairs 99% of the earthquakes both agencies report, with 98% of its pairs correct; the quality index ranks the offshore, high-gap population lowest, and a quality threshold removes 94% of it; and three preprocessing choices move the b -value by more than its formal error, namely a single global M_c from maximum curvature or the goodness-of-fit test on a catalogue whose completeness varies in space, a magnitude-selective quality filter (-0.06), and window declustering (-0.11). Above the more conservative b -value-stability estimate of M_c , which reaches the merged catalogue’s completeness, the estimator recovers the planted $b = 1.00$ of the independent events (1.01 ± 0.01). These figures characterise the controlled synthetic inputs and the software’s behaviour on them, not real New Zealand seismicity. The priority next steps are validation on heterogeneous real catalogues, measurement of duplicate-detection precision and recall, and calibration of the quality score against independent benchmarks. By making preprocessing choices explicit and exportable, CofC supports the shift towards reproducible, openly documented seismological practice advocated by the CORSSA community (Naylor et al., 2010) and the FAIR data principles (Wilkinson et al., 2016).

The platform is actively developed; community contributions, feature requests, and bug

reports are welcomed via the GitHub issue tracker.

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Data and Resources

The worked example in Section 7 uses *synthetic* seismicity data; no real event data are used or released. The two catalogues are generated by the seeded script `paper/figures/synthetic_catalogues.py` (Python; seed 42), whose parameters are listed in the electronic supplement (Table S1); the station sites are placed on New Zealand land from the Natural Earth 1:10m coastline. Everything the example reports is then computed by the platform’s own TypeScript modules, called as the platform calls them, through `paper/figures/worked_example_engine.ts`: the quality index on import (`lib/quality-scoring.ts`), duplicate association and the Quality-based merge (`lib/merge.ts`), and the M_c estimates (among them the b -value-stability estimate that sets the analysis cut-off), maximum-likelihood b -values and Gardner-Knopoff declustering (`lib/seismological-analysis.ts`). The only computations outside the platform are the symmetric-window variant of Gardner-Knopoff declustering, implemented in that file and labelled as such, and the scoring of the results against the generator’s ground truth. `paper/figures/generate_figures.py` runs the whole chain, draws Figures 4, 5 and 7, prints the summary and writes every quoted number to `paper/figures/worked_example_summary.json`; its Python dependencies are pinned in `paper/figures/requirements.txt`. The inputs of the construction are the true earthquake population (its zones, rates and planted b -values of 1.00 and 1.30, and the 23% aftershock share at $M \geq 2.0$), the two station networks and the recording and uncertainty model; the catalogue sizes, the overlap between the catalogues, and every quality score, duplicate count, M_c , b -value and declustering count are outputs. The continuous- integration suite exercises the same engine: `__tests__/paper/worked-example.test.ts` runs the complete pipeline on a small seeded two-catalogue input and checks its bookkeeping, and `__tests__/paper/merge-reproduction.test.ts` checks the matcher on a seeded construction with a 50% duplicate overlap. The GeoNet FDSN Event Web Service, from which users can obtain the live New Zealand catalogue, is available at <https://www.geonet.org.nz/>. The ISC-GEM catalogue ([Storchak et al., 2013](#)) is available at <https://www.isc.ac.uk/iscgem/>. Natural Earth map data are available at <https://www.naturalearthdata.com/>. All CofC source code and documentation URLs are given in Section 9. Additional screen captures of the platform’s analysis pages, using a reduced-scale version of the same synthetic catalogues, are provided in the electronic supplement.

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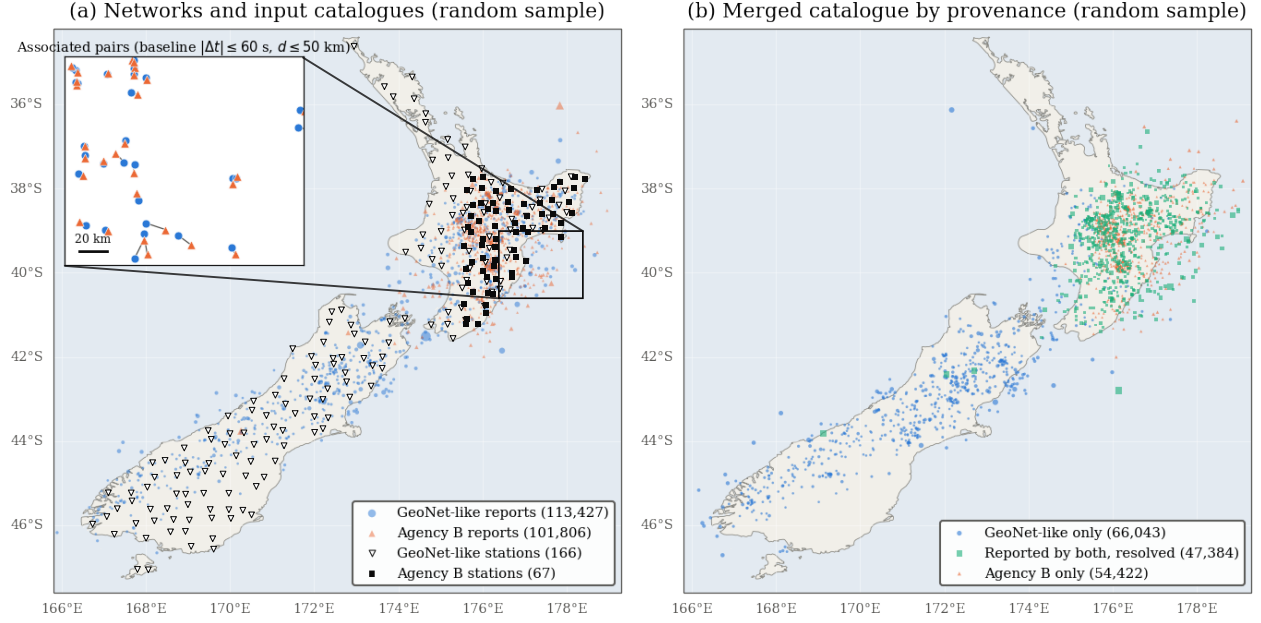
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Synthetic data for illustration; marker area increases with magnitude.

Figure 4: How the merge works, on the synthetic New Zealand example (Section 7). (a) The two synthetic station networks (GeoNet-like: open triangles; Agency B: filled squares) and a random sample of each agency’s entries (GeoNet-like: blue circles; Agency B: orange triangles). The inset zooms on the boxed region off the Hawke’s Bay coast and joins pairs of entries that the platform’s matcher associated as one earthquake (a random sample of 30 pairs), with a 20 km scale bar for comparison against the 50 km baseline distance window (equations (2)–(3)); magnitude and depth consistency are then checked as separate per-group gates. (b) A random sample of the merged catalogue coloured by provenance: events only the GeoNet-like network reported (blue circles), events both reported, resolved to a single solution (aqua squares), and events only Agency B reported (orange triangles), nearly all of them small events inside its denser regional network. Legend counts refer to the full catalogues. Marker area increases with magnitude. Data are synthetic and for illustration only.

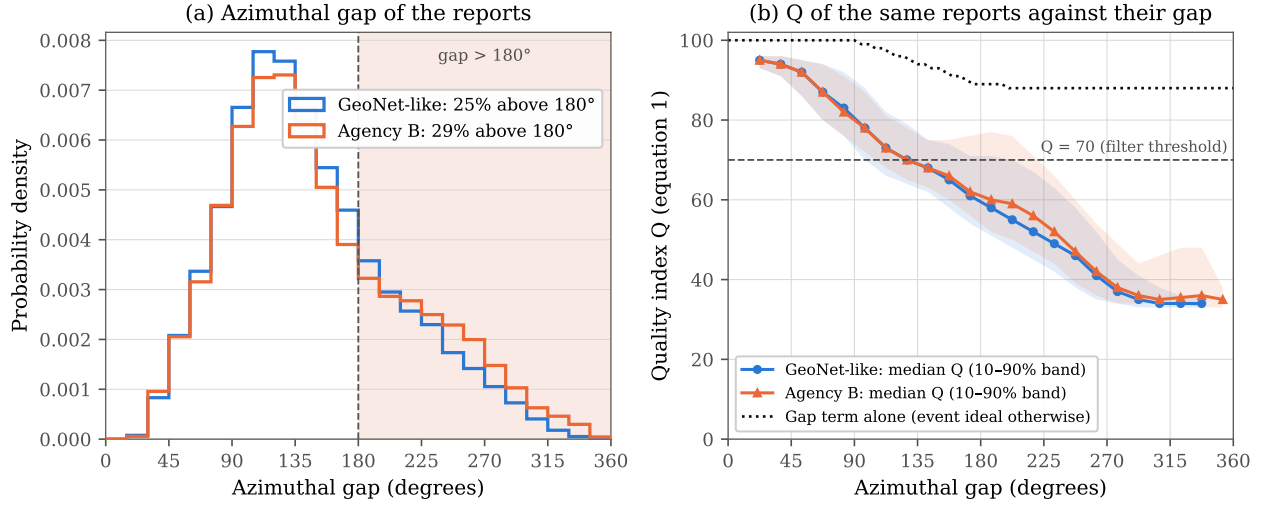


Figure 5: Azimuthal gap and the quality index on the synthetic example (Section 7). (a) Distribution of azimuthal gap for the GeoNet-like (blue) and Agency B (orange) entries before merging; the gaps are computed from each synthetic network’s geometry, so events offshore or outside a network have large gaps, and the shaded region marks gaps above 180°. (b) The quality index Q (equation (1)) that the platform assigns the same entries on import, against their gap: median and 10–90% range in 15° bins. The dotted line is the Q of an event that is ideal in every other respect: the gap term alone can lower Q by at most 12.5 points (to 88), so most of the fall comes from the larger location uncertainties and smaller station counts that the high-gap entries carry. The dashed line is the $Q \geq 70$ threshold applied in the example.

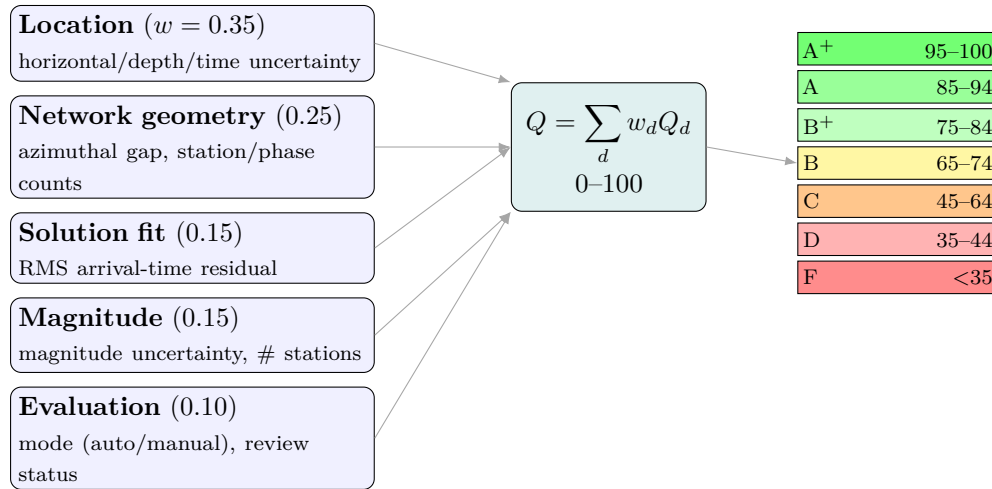


Figure 6: The per-event quality index (Section 3). Five dimensions, each computed from the listed event attributes and weighted as shown, are combined into a single score $Q \in [0, 100]$ (equation (1)) that maps to an A⁺–F grade (Table 3; colours match the map view). Missing attributes incur per-dimension penalties, so sparsely-documented events are scored conservatively.

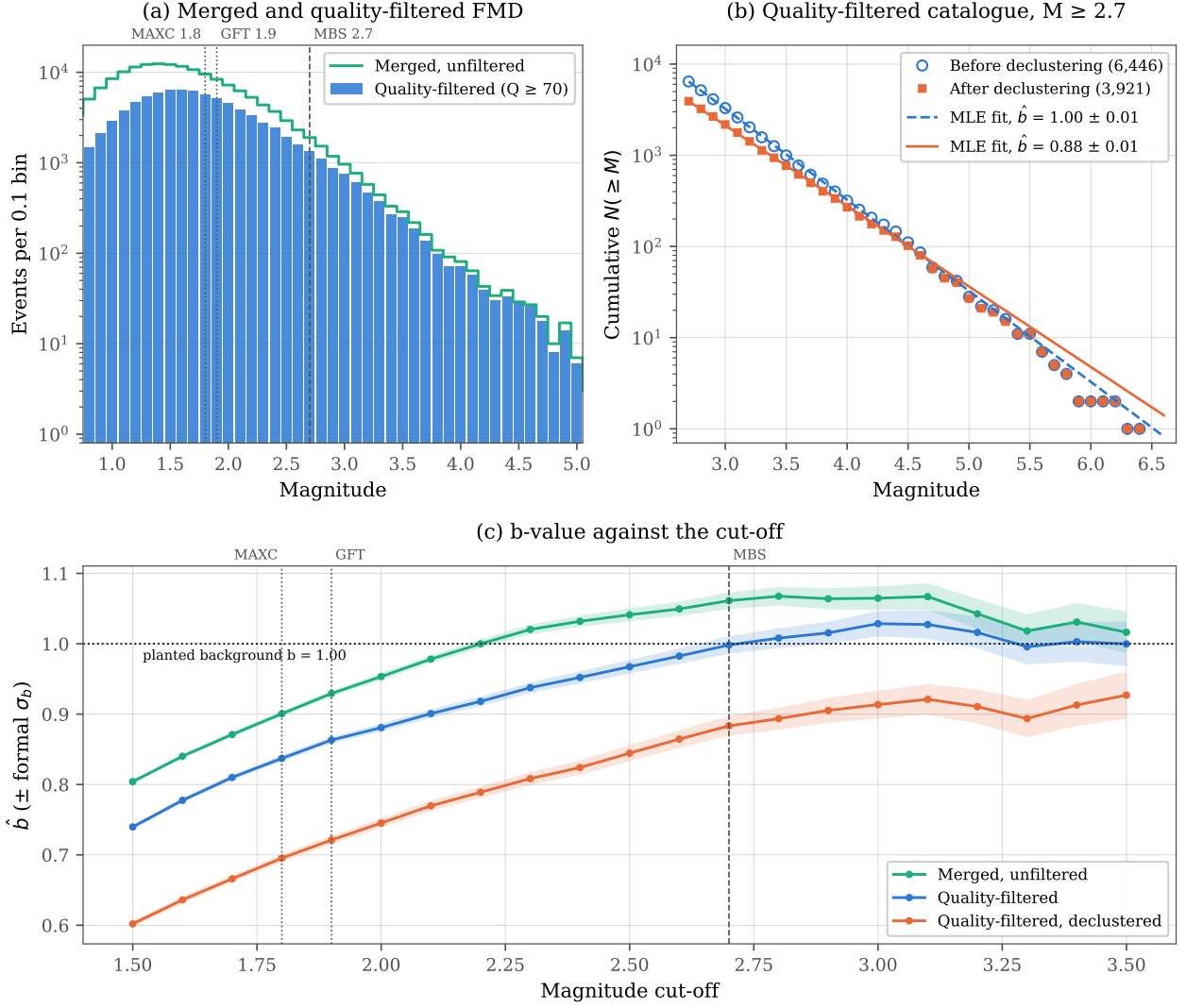


Figure 7: Completeness, b -value and declustering on the synthetic worked example, all computed by the platform (Section 7). (a) Non-cumulative frequency-magnitude distribution (FMD) of the merged catalogue (aqua outline) and of its quality-filtered subset ($Q \geq 70$, blue bars), in 0.1-wide bins centred on multiples of 0.1, with the platform’s MAXC and goodness-of-fit estimates of M_c for the filtered catalogue (dotted) and its b -value-stability (MBS) estimate for the merged catalogue, the cut-off from which $\hat{b}$ is stable (dashed); the filter removes a larger share of the small events. (b) Cumulative FMD of the 6 446 filtered events at or above $M_{2.7}$ before (open circles) and after (filled squares) Gardner-Knopoff declustering, with the platform’s maximum-likelihood fits; the partition is the one the windows of equations (11)–(12) produce on the events’ origin times and epicentres. (c) $\hat{b} \pm \sigma_b$ against the magnitude cut-off for the merged catalogue, the quality-filtered catalogue, and the quality-filtered catalogue declustered at each cut-off, with the M_c estimates of (a). $\hat{b}$ rises with the cut-off until the whole merged catalogue is complete, near $M_{2.7}$; the quality filter lowers it by about 0.06 and declustering by a further 0.11, below the planted background value (dotted). Data are synthetic and for illustration only.

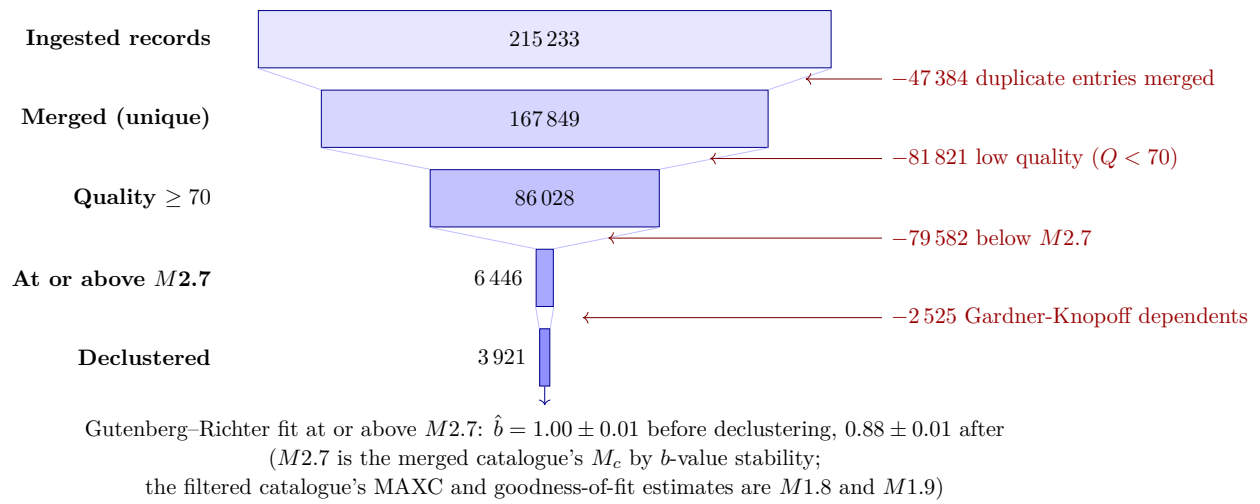


Figure 8: Event-count flow through the CofC pipeline for the synthetic worked example (Section 7), every count computed by the platform. Bar widths are proportional to the number of events; each stage removes the subset shown in red. The cut-off $M2.7$ is the platform’s b -value-stability estimate of M_c for the merged catalogue, the magnitude from which its b -value is stable (Figure 7c), well above the MAXC and goodness-of-fit estimates. All counts are properties of the controlled synthetic inputs, not real New Zealand seismicity.